

The Work a Verifier Needs

Transported summaries, certified low-precision heads
and the serving stack of a hybrid model

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Abstract

A speculative decoder asks its target model for a full vocabulary projection at every verified position, although a greedy verifier needs only a winner and a sampled verifier often needs only the truth of one inequality. We ask what work a verifier actually needs and test two sources of evidence that could let it do less: transporting a drafter’s tile summaries through the shared output head to the target state, as earlier revisions of this manuscript proposed, and evaluating the head at low precision with a rigorous per-row envelope, re-scoring only the rows that can still matter. The setting is Qwen3.5-4B in SGLang on one GH200, where the tied $248,320 \times 2,560$ head is 15% of the weight bytes of a plain decode step and 84% of those of a multi-token-prediction draft step (derived from the checkpoint).

We state the exactness contract a certified head can meet in a real engine: the token the stock head kernel returns at the same batch shape, certified only when a gap condition covers that kernel’s accumulation error and BF16 rounding, with the stock kernel as the fallback. We prove a floating-point envelope for the low-precision pass that holds for Hopper tensor-core accumulation under a published model, or needs no accumulation model for an integer pass; exact greedy selection and exact Gumbel-max sampling under a counter-keyed noise field from one refinement procedure; bounds on the rows that must be re-scored; and conditions under which exact sampling can finish before the scores are known. From the weights alone, transport’s envelope is narrower than an int8 envelope only if draft and target head inputs agree to within about 0.85% in relative ℓ_2 distance (median row, Euclidean envelopes, 64-row tiles); this compares envelope widths, not certification rates or runtime. The greedy certified head is prior art, and our claim is narrower (Section 2.8).

The measurements that decide the hypotheses (the drift of real draft states, candidate sets and head-path runtime, the attribution of time across the stack, and latency–throughput frontiers for an exact stack and for a lossy stack under a stated quality budget) are in progress. Each appears as a marked pending item, never as a number.

How to read the evidence in this revision. A *finding* is a theorem proved in the text or a result whose evidence is committed on the repository’s main branch. *Derived* numbers are calculations from the pinned checkpoint, configuration or source code; the formula and inputs are given where they are used. A number followed by [\[under review, PR #N\]](#) rests on evidence in an open pull request. [\[pending: ...\]](#) marks an experiment that has been designed but whose result is not yet committed, with the evidence path it will produce. Hypotheses are labelled as such. Table 2 gives the status of each question; Appendix F lists every claim with its evidence path.

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1 The question

Before choosing a faster implementation of an operation, it is worth asking which facts its consumer needs, and whether some of those facts are already implied by work done earlier. In speculative decoding the gap between what is computed and what is needed is concrete. At every verified position the target model projects its hidden state through the output head and produces a score for each of 248,320 tokens. A greedy verifier then asks for one winner. A sampled verifier asks whether a uniform draw falls below a ratio of probabilities, which is a single inequality. A tensor that is convenient to produce is not necessarily the right interface.

This paper asks what work a verifier actually needs, and tests two ways of supplying the missing facts more cheaply than a full projection. The first, which earlier revisions of this manuscript proposed and developed, reuses work the drafter has already paid for. A drafter that projects its own hidden state through the same head can summarize each vocabulary tile (its maximum score and its probability mass). When the target’s hidden state arrives, a Hölder bound transports those summaries to the target, and the verifier projects only the tiles whose bounds leave the decision open. We call this *transport*. The second source produces its own evidence: evaluate the head at low precision, attach a rigorous envelope to every row, and re-score in full precision only the rows whose interval can still change the decision. We call this *self-evidence*. Both keep an exact fallback, so neither changes the decision it certifies; they differ in what the evidence costs and in how tight it is.

The two mechanisms fail for different reasons. Transport’s error term for a row is the product of the row’s deviation from its tile centre and the change between draft and target hidden states. On a trained head both factors can be of the same order as the logits themselves, in which case the transported bound certifies nothing. Self-evidence pays for a full low-precision pass over the head, and its error term is the quantization residual times the hidden-state norm; the difficulty there is to make the envelope cover the floating-point arithmetic the engine actually performs, not an idealized dot product. Which failure dominates is an empirical question about a particular head and a particular engine, and answering it is the core of this revision.

The earlier revision was a research design with executable CPU references and no GPU evidence. This one tests the design in a real engine and reorganizes the paper around what the evidence shows. It keeps the mathematics that survives review, corrects what did not, and adds the theory needed to state exactness against a real kernel rather than against real arithmetic. It also widens the scope to the rest of the serving stack: when the head is only one term of the cost, a paper about avoided work has to say what the other terms are and what can be done about them.

1.1 The setting

The target is Qwen3.5-4B at revision 851bf6e806ef [57, 58]. Its configuration gives hidden width 2,560, a vocabulary of 248,320, and 32 decoder layers, of which 24 use Gated DeltaNet (GDN) linear attention [89] and 8 use full attention (every fourth layer). It has one native multi-token-prediction (MTP) layer, and the output head is tied to the input embedding. The checkpoint is multimodal; its vision tower is loaded by the engine but unused in every experiment here, which are text-only.

Table 1 gives the byte counts that frame the rest of the paper. They are derived from the checkpoint’s tensor sizes and the configuration, not measured. A plain decode step streams every decoder weight once plus the whole head; the embedding lookup reads only one row per request. The head is therefore 15.1% of the weight bytes of a plain step. An MTP draft step runs one small layer followed by the same head, so there the head is 84.1% of the weight bytes. The recurrent state is per request: the configuration stores it in FP32 (`mamba_ssm_dtype`), so each request holds $24 \times 32 \times 128 \times 128 \times 4$ bytes = 50.3 MB of GDN state, and a decode kernel that reads and writes it once per step moves 100.7 MB per request per step. That equals the weight bytes of a step at a batch of 84 requests. Section 4 replaces these calculations with measurements.

The engine is SGLang at commit bd66ce343e4f [97], with CUDA graphs and the overlap scheduler on, the FlashInfer attention backend [90] (FlashAttention 3 is not available on this aarch64 host), and the model revision pinned. The machine is one NVIDIA GH200 with 96 GB of HBM3 (compute capability 9.0) and a 64-core Grace

Component	Bytes	Derivation
Tied head (and input embedding)	1.271 GB	$248,320 \times 2,560 \times 2$ (BF16)
Decoder weights other than the embedding	7.140 GB	sum of tensor sizes in the checkpoint
Weights streamed per plain decode step	8.411 GB	decoder weights plus the head; head share 15.1%
MTP layer	0.241 GB	sum of tensor sizes; a draft step streams it plus the head (84.1% head)
Final decoder layer’s MLP	0.142 GB	$3 \times 2,560 \times 9,216 \times 2$
Vision tower (loaded, unused)	0.667 GB	sum of tensor sizes
GDN recurrent state per request	50.3 MB	$24 \times 32 \times 128 \times 128 \times 4$ (FP32)
Attention KV per context token	32 KiB	$8 \times 4 \times 256 \times 2 \times 2$ (BF16 K and V)

Table 1: Byte counts for Qwen3.5-4B at revision 851bf6e806ef, derived from the checkpoint’s safetensors headers and `config.json`. They are calculations, not measurements; Section 4 reports what the kernels actually move.

CPU. Two drafters are in scope. The native MTP layer is served through SGLang’s NEXTN path, which the engine reports as EAGLE [36]. The public DFlash block drafter for this exact target, `z-lab/Qwen3.5-4B-DFlash` [8, 92], has six layers, drafts blocks of up to 16 tokens conditioned on eight of the target’s intermediate hidden states, and has no head of its own: its drafts are projected through the target’s tied head. It is therefore both the strongest public speculative baseline for this model and a direct test bed for transport. The public DFlash 2 pair for Qwen3.8-27B [26, 27] is a transfer lane; no DFlash 2 checkpoint for the 4B model is public.

1.2 Hypotheses and their status

Table 2 lists the questions the work must answer, what would decide each, and where the evidence stands. The status column is updated as results are committed; a hypothesis that fails stays in the table with the reason.

1.3 Contributions

The contributions of this revision, at the strength the evidence currently supports, are these.

1. *An exactness contract for a certified head in a real engine* (Section 3). The reference is the stock head kernel’s decision for the same hidden state at the same batch shape, not a higher-precision recomputation. A gap condition (Theorem 3.5) states when a certified token must equal the stock token; every other row runs the stock kernel. The section also separates greedy, distributional and seeded contracts, states a general commit rule for decisions from partial information, and classifies skipped work as dead, deferred or soundly enclosed. These are proved results; the stock kernel’s error model on which the gap condition rests is an assumption that is tested, not proved.
2. *A comparison of the two evidence sources* (Section 5). Each envelope is the best its metadata allows (Proposition 5.3). Transport’s envelope is narrower than an int8 self-evidence envelope for row i only if the drift ratio $\rho = \|h^t - h^d\|_2 / \|h^t\|_2$ is below $\|e_i\|_2 / r_c(i)$ (Theorem 5.5). With 64-row tiles of contiguous token indices and the tile mean as centre, that threshold has a median of 0.85% over the rows of this head (weights only; a comparison of envelope widths, not of certification rates or runtime; `evidence/precision/head_constants.json`). The measured drift of real draft states is [\[pending: geometry\]](#).
3. *A floating-point envelope and the certified decisions built on it* (Sections 6 and 7). The envelope covers Hopper tensor-core accumulation under the published model of Khattak and Mikaitis [28], or needs no accumulation model when the pass is an integer product. One refinement procedure gives exact greedy selection and exact Gumbel-max sampling under a counter-keyed noise field, with a bound on the expected number of re-scored rows. Fixed-noise verification [12], with the stock seeded sampler’s noise field, turns sampled verification into a certified argmax with no partition function; it is the sampling contract adopted for the certified speculative

	Hypothesis	Deciding evidence	Status
H1	The head is a large share of decode and MTP-speculation time for this model on GH200.	nsys attribution at several batch sizes (Section 4).	Byte shares derived (Table 1); measured shares [pending: profile, PR #13]
H2	Transport gives useful certificates on real hidden states.	Aligned MTP and DFlash replay: bound widths against logit margins (Section 5).	Crossover derived from weights; measured drift [pending: geometry]
H3	A low-precision head with a rigorous envelope certifies the stock decision while reading fewer weight bytes.	Replay candidate sets, then a GPU kernel against the stock head chain (Section 6).	Envelope and contract proved; candidate sets and runtime [pending: geometry, kernel]
H4	The certificate improves the served latency–throughput frontier without changing outputs.	Before/after Pareto sweeps with output-equality checks (Section 11).	[pending: bench, integration]
H5	SGLang’s hybrid state stays correct under speculative verification.	Differential tests against plain decoding at every rejection position, aborts and prefix reuse (Section 3.8).	[pending: state]
H6	Other layers of the stack leave measurable headroom.	Profiles and controlled ablations (Sections 4, 10).	[pending: profile, moonshot]
H7	Reformulations and declared approximations multiply gains well beyond tuning.	Bytes per step, quality–speed curves, end-to-end sweeps (Section 10).	Ceilings derived; measurements [pending: moonshot]
P1	A certificate on the decision can skip the final layer’s MLP and the head in target-only decoding.	Oracle-radius replay at the cut, then real bounds (Section 6.7).	Hypothesis; kill test [pending]
P2	Long-window exact repair from a DFlash window beats optimized DFlash substantially.	Perfect-continuation oracle, then real proposals (Section 9).	Hypothesis [pending: repair]
P3	Target-anchored residual repair is much cheaper than recomputation.	Oracle and real-correction kill tests (Section 9).	Hypothesis [pending: repair]
P4	Writing the recurrent state less often, bit-exactly, gives at least 1.10× decode at batch 128.	Bit-exactness check, then paired timing (Section 10).	Hypothesis [pending: moonshot]
P5	Exact computation can be shared across unrelated requests.	Analysis and exact witnesses; measured ceiling of the one structural candidate (Section 10).	Rejected in analysis; ceiling [pending: profile]

Table 2: Questions and the evidence that decides them. H1–H7 are the working hypotheses of the programme; P1–P5 are later proposals with designed kill tests or pre-registered criteria.

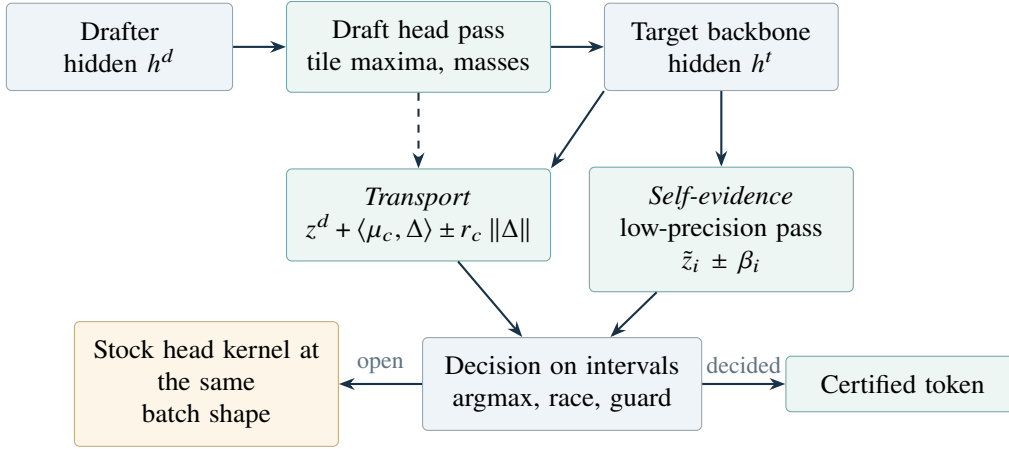


Figure 1: Two sources of evidence for the same decision. Transport reuses the drafter’s head pass and bounds only the change $\Delta = h^t - h^d$; self-evidence pays for a cheap pass over the head at the target state. Either way the decision is taken on intervals, rows or tiles are refined while it stays open, and a row that cannot be certified runs the stock head kernel. The target backbone always runs; the saving, if any, is in the head.

path (design decided, measurement pending). Acceptance guards with enclosed numerators, residual races, a characterization of how much of an exact sample can be emitted before the scores are known, and KL and total-variation bounds for an approximate mode complete the set. Several of these rest on known principles (Gumbel-max, A* Sampling [43], maximal coupling); Section 2 says which parts are new.

4. *Measurements on every axis of the serving stack* (Sections 4 and 8–11): attribution and bytes per step, drafting, the certified head’s runtime, repair, recurrent state, and two latency–throughput frontiers, one exact and one lossy under a stated quality budget. These are [pending] in this revision.

The greedy form of the certified low-precision head is not new: several public implementations screen a low-precision copy of an LM head with a bound and re-score the survivors exactly (Section 2.3). Nor is using bounds to obtain an exact Gumbel-max sample [43, 45]. What we claim is narrower, and Section 2.8 states it.

1.4 Reading guide

Section 2 places the work against prior art. Section 3 fixes what “exact” means in this engine. Section 4 measures where the time goes. Sections 5–7 develop and test the two evidence sources and the decisions built on them. Sections 8–10 cover drafting, repair and the rest of the stack, and Section 11 reports the served frontiers. Section 12 collects what did not work and the limits of what did. The appendices hold the engine integration notes, the CPU evidence inherited from the earlier revision, mechanisms not yet tested, reproduction commands, the claim ledger and the attribution statement.

2 Where prior work already reaches

A source audit should remove false novelty before experiments make it expensive to abandon. This section is organized by the mechanism each body of work shares with ours, and it ends with the narrow claim that remains. Descriptions come from the literature review that accompanies this revision (sources/literature_review.md); performance figures quoted from other work are the authors’ own and were not reproduced.

2.1 The head as a cost

For a small model with a large vocabulary the output head is a large part of each step. FR-Spec profiles EAGLE-2 drafting on Llama-3-8B, with a 128k vocabulary, and finds the head projection at 49% of draft time and the softmax over its output at a further 13% [96]; a trimmed-vocabulary study finds the head at 64% of an EAGLE-3-style

drafter’s FLOPs [3]. Vocabulary sizes have grown with compute budgets [72], which is why 4B models now ship 250k-row heads. Table 1 gives the corresponding byte shares for Qwen3.5-4B; Section 4 measures the time shares.

2.2 Screening, exact search and bounds

Computing cheap scores for every row, bounding the error, and re-computing only the rows that could still matter is an old idea in similarity search. The VA-file filters with quantized cell approximations and refines survivors exactly [83]; FEXIPRO combines an SVD transform with scaled-integer upper bounds and exact floating-point refinement [35]; LEMP prunes norm buckets with length and Cauchy–Schwarz bounds [73]; ball and cone trees bound $\max\langle q, p \rangle$ by $\langle q, \mu \rangle + R \|q\|$ [60]. These methods are exact in real arithmetic. A warning comes with them: Maximus and Optimus found that blocked dense matrix multiplication often beats such indices end to end [2], so a pruning method must beat a tuned dense kernel, not a count of floating-point operations.

In language models, SVD-Softmax computes preview logits from leading singular directions and recomputes a fixed top set in full [68], and HiRE uses an int4 or low-rank copy of the head to choose a fixed top- k' set that is recomputed in BF16 [32]. Both have the two-pass shape of self-evidence without a certificate: a fixed candidate count carries no guarantee that the true winner is among the candidates. CSV-Decode clusters the head’s rows offline and bounds each cluster by $\langle \mu_c, h \rangle + R_c \|h\|$, opening clusters until the top- k is certified [40]. That is exactly the static tile screen that earlier revisions of this manuscript proposed as an ablation, and it is exact in real arithmetic; its text does not treat floating-point rounding. A public experiment on a Qwen3.8 head of the same vocabulary found that centroid-radius bounds prune almost nothing there [59]. The transport bound differs from these screens only in where it starts: from a score already measured at the draft state, bounding the change $\langle w_i - \mu_c, \Delta \rangle$ rather than the score itself. Bounding an inner product under a bounded displacement of the query is also known, for example in sequential screening rules for the lasso [78] and in cached retrieval with drifting queries [16].

2.3 Certified low-precision heads

The greedy half of the self-evidence mechanism is implemented publicly, several times over.

- sparkpipe keeps an E4M3 shadow of a BF16 head in 32-element groups, attaches a per-group certificate that includes γ_n floating-point terms for CUDA-core FMA accumulation, and re-scores survivors exactly in BF16. It routes the decode, last prefill and MTP argmax rows of Qwen3.8-27B (a 248,320-row head) through this path and reports bit-exact tokens and scores [69]. Sampled requests take the full head read.
- dgpp stores an int8 head as per-row bit planes, bounds every logit from the high planes, and re-reads only surviving rows in the production accumulation order, for MTP draft steps and greedy verification [22]. Its reference is the quantized head itself, and its floating-point term is a fixed relative slack.
- A Metal implementation for Apple silicon screens with an MXFP8 coarse head whose bound covers quantization and both kernels’ rounding, and re-scores candidates with a textual replica of the stock BF16 kernel so that rows are bit-identical [33].
- knlp certifies greedy decoding with an int8 shadow in a PCA basis and a Cauchy–Schwarz bound on the out-of-basis part; its own table reports an argmax match of 0.998 at 14B, so floating point is not fully handled [6].

In retrieval, EAHR bounds int8 inner products by $\|e_q\| \|v\| + \|\hat{q}\| \|e_v\|$ with a floating-point guard and rechecks ambiguous items in FP32 [93]. Du et al. show why the reference must be named: BF16 and FP16 greedy decoding diverge on 49–100% of prompts in their experiments, a flip is decided at the head by the top-two margin, and recomputing in FP32 when the margin is small (a heuristic threshold, not a certificate) repairs much of it [14]. Measured head sensitivity sets expectations for re-scoring: a row-wise int8 copy of the Qwen3-8B head agrees with the BF16 argmax on 97.56% of positions [29], so an exact re-score must catch the rest.

2.4 Exact sampling, fused sampling and determinism

The Gumbel-max identity [21, 54] makes a categorical sample the argmax of perturbed scores, and its top- k form samples without replacement [30]. A* Sampling uses bounds to instantiate only the relevant perturbations and still returns an exact sample [43]. Mussmann, Levy and Ermon draw Gumbels only for a retrieved top set, bound every other score by the smallest retrieved one, and lazily instantiate the few tail perturbations that could win; with an approximate top set they widen the cutoff by the approximation error [44, 45]. That is the direct precedent for exact Gumbel-max sampling from bounded scores. The version used here differs in two engineering respects: a dense low-precision pass bounds every row instead of a sublinear index, and the noise is a fixed field keyed by token, so the bounded race returns the same token as a dense race with the same field, not only a sample with the right law.

FlashSampling fuses Gumbel-max sampling into the head projection, keeping one perturbed maximizer per row and tile so logits never reach memory; it computes every tile, accumulates in FP32 and is exact in distribution [62]. Its noise offset depends on the tiling, so a token’s noise cannot be recomputed on its own. FlashInfer samples top- k , top- p and min- p without sorting and implements chain speculative sampling [87, 90]. SonicSampler fuses the post-logit sampling and verification kernels over a bounded top-128 pool and describes top- p under that pool as theoretically lossy [55]. On determinism, batch-invariant kernels make temperature-0 outputs reproducible across batch compositions [23]; SGLang’s deterministic mode replaces the default matrix multiplication with a persistent Triton kernel and forces the PyTorch sampler [74], which changes the kernel a certificate would have to reproduce. Floating-point nondeterminism in inference has also been characterized and partly mitigated [91].

2.5 Speculative decoding and verification policies

Speculative sampling accepts a proposal $x \sim q$ with probability $\min(1, p_x/q_x)$ and otherwise samples the normalized residual $(p - q)_+$; the output has the target law [7, 34], and the acceptance probability $\sum_i \min(p_i, q_i)$ is that of a maximal coupling. SpecTr and block verification extend the rule to several drafts and to whole blocks [70, 71]. Drafter-invariant speculative decoding couples draft and target through shared Gumbel noise, so the committed token is $\arg \max_i (\log p_i + G_i)$ whatever the drafter proposes [12], and the construction extends to several drafts [61]. Under that coupling sampled verification is an argmax, which is why it matters here (Section 7.3).

Drafters that project through a head pay for it at every draft step: Medusa and the EAGLE family [4, 36–38], native multi-token prediction [13, 19], and DFlash’s block-diffusion drafter [8], whose public 4B checkpoint drafts through the target’s head [92]. DFlash 2 adds a context-gated bilinear selector over candidate lattices [26, 27]; LiLiCorr reranks DFlash candidates and finds that an exact best-path search over a surrogate score decodes worse than the greedy rule [64]. The model card of the 4B DFlash drafter reports, on one B200 with greedy decoding, 3.40–4.60 $\times$ over autoregressive decoding at concurrency 1 with blocks of 16 and 2.15–2.61 $\times$ at concurrency 32 with blocks of 8, against at most 2.31 $\times$ and 1.80 $\times$ for the MTP layer [92]. These are the authors’ numbers on different hardware; Section 8 measures the drafter here.

Verification-depth policies choose how many drafted tokens to verify. OPT-Tree searches for the tree that maximizes expected acceptance [79]; Sequoia sizes trees for the hardware [10]; SpecDec++ stops drafting with a learned acceptance head [25]. DSpark schedules verification from confidences and a profiled throughput curve [11], D-cut prunes draft blocks against a profiled cost table aligned with the engine’s CUDA-graph shapes [41], and an open SGLang pull request ports the DSpark scheduler to DFlash 2 with greedy benchmarks [66]. For sampled decoding, DSpark states that lossless speculation requires admission decisions that do not depend on future candidate tokens, with a counterexample, and D-cut gives a two-token counterexample and a shifted-confidence repair [11, 41]. The counterexample in Section 7.8 illustrates their condition; it is not a contribution of this paper.

2.6 Draft-vocabulary reduction

FR-Spec restricts the draft softmax to a frequency-ranked subset of the head [96], VocabTrim trims by token counts on target-generated text [20], SpecVocab selects a per-step subset with low-rank scores and computes

Existing work	Already established or implemented	Question pursued here
CSV-Decode [40]; ball bounds [60]	Cluster centre-plus-radius bounds on head rows; exact top- k in real arithmetic.	Transport starts from draft scores; measure it against this static screen and against self-evidence.
sparkpipe, dgpp, Laguna prune, knlp [6, 22, 33, 69]	Certified greedy screening with a low-precision head and exact re-score, including MTP rows of a 248,320-row head.	Hopper-sound envelope; stock-kernel contract; exact sampling; SGLang measurement.
HiRE, SVD- Softmax [32, 68]	Cheap copy selects a fixed candidate set; exact recompute.	Certified, variable candidate sets; fixed- k' as an ablation.
A* Sampling; lazy Gumbels [43, 45]	Exact samples from bounded scores.	A dense cheap pass and a keyed noise field for pathwise equality with the dense race.
FlashSampling [62]	Gumbel-max fused into the head; every tile computed.	Skip rows, not just logit materialization; dense comparator.
Drafter-invariant decoding [12, 61]	Shared-noise verification; output independent of the drafter.	Certified argmax for sampled verification with no partition function.
DSpark, D-cut, PR 36136 [11, 41, 66]	Confidence- and cost-scheduled verification depth; the non-anticipating condition.	Depth policies under fixed-noise verification, where any policy is exact.
DFlash, DFlash 2 [8, 26, 92]	Block drafting through the target head; public 4B DFlash drafter.	Strongest speculative baseline here; test bed for transport.
ReplaySSM; GDN tree verification [18, 39, 65, 81]	Replay-based rollback; tree verification for GDN hybrids.	Measured state traffic and capacity; state as an axis of the stack.

Table 3: Prior art bounds the novelty claim. Reported performance of other work is not reproduced by this paper.

exact draft logits on it [84], DynaSpec routes contexts to clusters of rows [94], and NanoSpec (titled MicroSpec in its first version) builds a small in-context vocabulary without training [9]. FastMTP compresses only the draft vocabulary and keeps full-vocabulary verification [5]. All of these change only the drafter, so the target’s decision is unchanged when the verifier uses the drafter’s actual proposal law, or when verification is target-only. They trade acceptance for draft cost; a certified head returns the full-vocabulary draft token while reading fewer bytes per row, and it also applies on the target side, where truncation does not.

2.7 Hybrid recurrent state under speculation

Gated DeltaNet’s recurrence [88, 89] makes rollback after a rejection a state problem. The Mamba-in-Llama work advances one cached state lazily and recomputes after a rejection [80]; STree, TreeWY, Bole, GDN Tree-Scan and SpecLA verify trees or chains on SSM and GDN hybrids [18, 42, 81, 82, 86], and TreeWY says explicitly that its outputs are not bit-identical to the baseline. ReplaySSM caches recent inputs instead of writing the state at every step, equivalent to ordinary decoding up to floating-point error [39]; SGLang has its own replay fold behind flags that are off by default [65]. State precision matters for quality: DAMP measures FP16 recurrent state as nearly lossless and BF16 as several points worse on a reasoning benchmark for another GDN model [95], and LeapQuant re-quantizes the state once per 16-token window [53].

2.8 The novelty boundary

We do not claim norm or quantization bounds, bound-then-refine search, certified greedy screening of an LM head, top- k merging, Gumbel-max sampling from bounded scores, drafter-invariant verification, the non-anticipating condition for adaptive depth, function scans, recurrent replay or graph-aware batching. Table 3 lists the closest work and the question left for this programme. What remains, as far as the review found as of 30 September 2026 with the coverage limits it records, is:

1. a floating-point envelope for a low-precision head pass that is sound for the accumulation the GPU actually performs (NVIDIA tensor cores align adds to the largest term and truncate [15], and Khattak and Mikaitis model Hopper’s BF16 and FP8 paths in detail [28]; existing certified heads use CUDA-core FMA, replay the production order, or assume FP32 is exact);
2. an exactness contract against the stock kernel at the same batch shape, with a stated gap condition and fallback;
3. extension of the certified head from greedy selection to exact sampling under a fixed counter-keyed noise field (pathwise equal to the dense race) and to fixed-noise verification with the engine’s own seeded noise, which keeps seeded outputs token-identical to non-speculative decoding; partition brackets for standard acceptance are analysed but not built;
4. a measurement inside SGLang on HBM3 with CUDA graphs, applied to draft steps, verification and plain decoding, against the stock head chain and FlashSampling;
5. for transport, the only mechanism not found in prior work (draft-side tile summaries carried through a shared head to prune the verifier’s head), a measurement of whether it survives real geometry.

Items 1–3 are modest extensions of known principles; their value depends on measurement. The contribution is mostly soundness, integration and measurement, and a well-explained negative result would also be informative.

3 Exactness in a real engine

A certificate is a statement about a specified computation. “Exact” is therefore incomplete until it names the computation whose output is preserved and the sense in which it is preserved. This section fixes both for the engine used here. It separates the contracts a decoder can promise, describes what the stock engine actually computes, states a general rule for committing a decision from partial information, and derives the condition under which a certified token equals the token the stock head kernel would have produced.

3.1 Three contracts, and why the reference is an implementation

Three contracts need to be kept apart. *Greedy equivalence* requires the same token as the reference’s argmax at every reached prefix, with the same logit processors and the same tie rule. *Distributional equivalence* requires the same conditional law of the next token at every reached prefix. *Seeded equality* requires the same token for the same seed, which fixes the sampler and its random-number coupling as well as the law. None implies the next. Logits (1, 0) and (2, 0) have the same greedy decision but different laws at temperature one (0.731 against 0.881 on the first token). Two samplers with the same law generally produce different text under the same seed. And equal token sequences say nothing about latency or about whether the cache that produced them is owned correctly.

The reference for each contract is an implementation, not an idealized network. Mathematical equivalence over the reals does not imply that two GPU reductions choose the same token, because changing the batch shape or reduction order can move logits that are close [49, 56]. A certificate must therefore enclose the value the reference implementation computes, including its casts, and not only the real-arithmetic dot product. A deployment may admit a weaker contract, but it must name it and evaluate against it rather than call it bitwise exact.

For sampled verification, let p be the target law at the reached prefix and q the *actual* proposal law, conditional on the information used to produce the proposal. A softmax buffer from a different selector is not q ; a deterministic proposal is a point mass. The standard one-step correction accepts $X \sim q$ with probability $\min(1, p_X/q_X)$ and otherwise samples $(p - q)_+$ normalized; the accepted mass at i is $\min(p_i, q_i)$ and the residual contributes $(p_i - q_i)_+$, which sum to p_i [34]. The certificates in this paper change how such a comparison is evaluated, never the law used in it.

3.2 What the stock engine computes

At the pinned commit, with the defaults used here, the head is `torch.matmul(h.to(bf16), W.T)` in `LogitsProcessor._compute_lm_head` (`python/sglang/srt/layers/logits_processor.py`). The

product is a cuBLAS BF16 GEMM that accumulates in FP32 and returns BF16 logits, which are then copied into an FP32 buffer. cuBLAS chooses its algorithm per shape, so the logits of a row can depend on how many rows are in the batch, and PyTorch’s default permits reduced-precision reductions in BF16 GEMMs. Greedy decoding and greedy verification take `torch.argmax`, which returns the first maximal index. BF16 keeps eight significant bits, so logits in $[16, 32)$ are spaced 0.125 apart; two real logits closer than about one spacing can round to the same BF16 value, and then the smaller token index wins.

Sampling at the pin is already a race. Without $\text{top-}k$, $\text{top-}p$ or $\text{min-}p$ and without a seed, the sampler returns $\arg \max_i p_i / E_i$ with E_i exponential draws from the Philox generator, which is Gumbel-max in another form (SGLANG_OPT_USE_GUMBEL_SAMPLE, on by default). With a per-request seed it returns $\arg \max_i (\log p_i + G_i)$ in FP64, where $G_i = -\log(-\log x_i)$ and x_i is a 32-bit MurmurHash of (seed, position, token) scaled to $[0, 1]$ (multinomial_with_seed in `python/sglang/srt/layers/sampler.py`). That is a counter-based noise field: the noise of any token can be recomputed on its own, in any order. Sampled speculative verification by default uses point-mass drafts: a drafted token is accepted when a uniform coin falls below the target probability mass of the siblings tried so far, and a rejection samples the target restricted to the untried tokens by inverse CDF. Two flags change the reference: `--enable-fp32-lm-head` makes the head return FP32 logits, and `--enable-deterministic-inference` replaces the default matrix multiplication with a persistent batch-invariant Triton kernel [74].

3.3 A decision needs a sound set, not a finished forward pass

The same rule underlies every certificate in this paper. Partial computation yields information I about the final logits: an approximate pass with error bounds, a transported summary, an approximate hidden state with a radius. From I we keep a *sound set* $Z(I) \subseteq \mathbb{R}^V$ that contains the reference implementation’s logits. A decision rule $D(z, R)$ maps logits and the randomness fixed by the contract (a noise field, a uniform draw) to a token or an accept bit.

Definition 3.1 (Commit rule). Commit the decision d when $\{D(z, R) : z \in Z(I)\} = \{d\}$. Otherwise acquire more information, which must shrink or replace $Z(I)$ by another sound set, and test again.

If the reference logits z^* lie in $Z(I)$, a committed d equals $D(z^*, R)$; that is the whole soundness argument, and it holds whatever produced $Z(I)$. For a box $Z(I) = \prod_j [l_j, u_j]$ the greedy rule commits c when $l_c > \max_{j \neq c} u_j$. A Gumbel race with a fixed field G commits c when $l_c/T + G_c > \max_{j \neq c} (u_j/T + G_j)$, which needs no partition function; the field must stay fixed through every refinement [43]. When the information is about the hidden state rather than the logits, a box is loose, and the directional form below is the right certificate.

Proposition 3.2 (Directional margin). *Let the logits be $z = Wh$ in real arithmetic and suppose $\|h - \tilde{h}\|_2 \leq \varepsilon_h$ for a known $\tilde{h}$. Then c is the strict argmax of Wh for every such h if and only if*

$$(w_c - w_j)^\top \tilde{h} > \|w_c - w_j\|_2 \varepsilon_h \quad \text{for all } j \neq c.$$

This condition is implied by, and can be much weaker than, the box condition obtained from the row bounds $|z_j - w_j^\top \tilde{h}| \leq \|w_j\|_2 \varepsilon_h$.

Proof. For fixed j , $\min_{\|e\|_2 \leq \varepsilon_h} (w_c - w_j)^\top (\tilde{h} + e) = (w_c - w_j)^\top \tilde{h} - \|w_c - w_j\|_2 \varepsilon_h$ by Cauchy–Schwarz, with equality at $e = -\varepsilon_h (w_c - w_j) / \|w_c - w_j\|_2$. So $z_c > z_j$ throughout the ball exactly when the stated inequality holds. The box condition reads $(w_c - w_j)^\top \tilde{h} > (\|w_c\|_2 + \|w_j\|_2) \varepsilon_h$, and $\|w_c - w_j\|_2 \leq \|w_c\|_2 + \|w_j\|_2$. $\square$

The directional certificate is not free: it needs $\|w_c - w_j\|_2$ for every competitor, which for all j costs as much as a head pass. It is useful after a box screen has reduced the competitors to a few rows. It is also a real-arithmetic statement; Section 6 supplies the floating-point terms that turn such statements into enclosures of computed values.

3.4 Contracts for the head and agreement with the stock kernel

Let $\mathbb{B}$ be the BF16 numbers. The stored head is $W \in \mathbb{B}^{V \times D}$ and $h \in \mathbb{B}^D$ is the vector the engine feeds to it. The *real logit* is $z_i = \sum_j w_{ij} h_j$ in exact arithmetic. A binary format with precision p (significant bits) and smallest normal exponent $e_{\min}$ has unit roundoff $u = 2^{-p}$ (BF16 $p = 8$, FP32 $p = 24$, FP64 $p = 53$). Round to nearest satisfies $|\text{fl}(x) - x| \leq u|x| + \mu$ with $\mu = 2^{e_{\min}}$, which also covers flush-to-zero; directed and truncating rounding satisfy the same bound with $2u$. Write rn_B for rounding to nearest-even BF16 and $\text{ulp}_B(x)$ for the BF16 spacing at $|x|$. The stock head returns $L_i = \text{rn}_B(\hat{y}_i)$, where $\hat{y}_i$ is cuBLAS’s FP32 accumulation of the products.

Definition 3.3 (Greedy and race contracts). **R-real**: $\text{tok} = \min\{i : z_i = \max_j z_j\}$. **R-bf16**: $\text{tok} = \min\{i : \text{rn}_B(z_i) = \max_j \text{rn}_B(z_j)\}$. **R-stock**: $\text{tok} = \min\{i : L_i = \max_j L_j\}$. **S-race**: for $T > 0$ and a noise vector $g \in \mathbb{R}^V$, $\text{tok} = \min\{i : z_i/T + g_i = \max_j (z_j/T + g_j)\}$.

R-real and R-bf16 are functions of (W, h) alone, so they do not depend on reduction order or batch shape. R-stock is what an unmodified server returns, and it can depend on both.

Lemma 3.4 (BF16 separation). *If $y_a - y_b > \text{ulp}_B(\max(|y_a|, |y_b|))$ then $\text{rn}_B(y_a) > \text{rn}_B(y_b)$.*

Proof. Rounding is nondecreasing. If both round to v , both lie in the preimage of v . When $|v|$ is not a power of two the preimage has length $\text{ulp}_B(v)$ and both values lie in the binade of v . When $|v| = 2^e$ the preimage is $[v - \text{ulp}_B(v)/4, v + \text{ulp}_B(v)/2]$ (for $v > 0$); two values on one side are within half the spacing of their binade, and values on both sides are within $\frac{3}{4} \text{ulp}_B(v) \leq \text{ulp}_B(y_a)$. At $v = 0$ the preimage has the subnormal spacing. Each case contradicts the hypothesis. $\square$

Theorem 3.5 (Agreement with the stock head). *Let $|\hat{y}_i - z_i| \leq G_i$ for every i and let a be the R-real token. If for all $b \neq a$*

$$z_a - z_b > \delta_{ab} := G_a + G_b + \text{ulp}_B(\max(|z_a|, |z_b|) + \max(G_a, G_b)),$$

then R-stock returns a . If $\text{rn}_B(\hat{y}_i) = \text{rn}_B(z_i)$ for all i , R-stock equals R-bf16.

Proof. $\hat{y}_a - \hat{y}_b > \text{ulp}_B(\max(|z_a|, |z_b|) + \max(G_a, G_b)) \geq \text{ulp}_B(\max(|\hat{y}_a|, |\hat{y}_b|))$, so Lemma 3.4 gives $L_a > L_b$ for every b . The second claim is immediate. $\square$

The contract used in this paper. A certified head claims R-stock at the batch shape it serves: for each row it returns the token the stock head kernel would return for the same hidden state in a batch of the same shape. It certifies a row only when the gap condition of Theorem 3.5 holds, with G_i bounding the stock kernel’s pre-rounding error at that shape; for every other row it runs the stock kernel on the batch and takes that kernel’s own argmax and tie rule, so those rows are the stock output by construction. Near $|z| \in [16, 32)$ the margin required is about $0.125 + 2G$.

The condition rests on G_i , and G_i rests on a model of the stock kernel. For an FP32 accumulation, Lemma 6.1 supplies G_i from the reduction tree; for the tensor-core kernels cuBLAS selects on this GPU, the tree’s per-node error comes from the published model of Hopper accumulation [28], which is not a vendor contract, and a reduced-precision split- K reduction admits no documented bound at all. PyTorch permits such reductions by default, so either the stock baseline disables them or the fallback must cover every shape at which cuBLAS might use them; the exact reference contains a constructed witness in which BF16 partial sums reverse a token that the FP32-model gap condition certifies (`tests/test_precision.py`, `evidence/precision/`). The agreement of certified tokens with R-stock is therefore conditional on the stock kernel’s error being within the modelled G_i at the shapes served, and that assumption is tested against the kernel’s outputs rather than proved [pending: kernel: measured cuBLAS error against the model at the served shapes, and certified versus stock token agreement on captured hidden states; `evidence/certified_head/`].

R-bf16 is not the contract, although it is attractive: it is batch-invariant and it uses the same lowest-index rule as the stock argmax. It is not what the engine computes. The stock kernel rounds its own FP32 accumulation, which

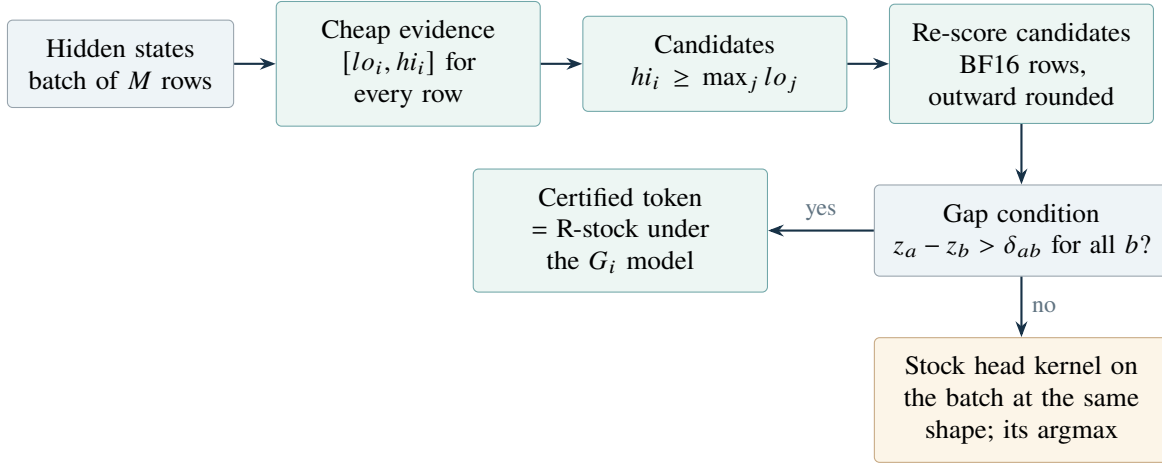


Figure 2: The exactness contract. Cheap evidence (self-evidence or transport) and re-scoring decide R-real for most rows; the gap condition of Theorem 3.5 converts that decision into the stock kernel’s decision when the margin covers the stock kernel’s modelled accumulation error and BF16 rounding. A row that fails runs the stock kernel at the same batch shape, so its token is stock by construction. The kernel implementation is described in Section 3.5.

can fall on the other side of a BF16 rounding boundary from the exactly rounded value. A two-row witness shows it: with $w_0 = (16, 0, 0)$, $w_1 = (16, \frac{1}{16}, 2^{-30})$ and $h = (1, 1, 1)$, the exact logits are 16 and $16.0625 + 2^{-30}$, so R-bf16 rounds them to 16 and 16.125 and picks token 1; a sequential FP32 accumulation loses the 2^{-30} , rounds 16.0625 to even, returns 16 for both rows, and the stock argmax picks token 0 (`test_bf16_contract_is_not_stock` in `tests/test_precision.py`; the gap condition refuses to certify this row). Using R-bf16 as the reference would report agreement with a computation the engine does not perform. R-real remains useful as a batch-invariant reference for replay studies, and the FP32-logit contract equals the engine run with `--enable-fp32-lm-head`; both are named as such wherever they are used.

3.5 How the fallback runs at the same batch shape

[pending: kernel: what the certified head’s stock-reference mode guarantees; the assumed cuBLAS error model and its validation against measured cuBLAS error; how the dense fallback is captured as a conditional node inside the CUDA graph so that it runs on the whole batch at the served shape; the undecided rate and its causes; draft in `~/vp-coord/drafts/kernel.tex`]

3.6 Every divergence has a mechanism

Stock configurations that ought to be equivalent already disagree: plain decoding at different batch sizes, plain decoding against MTP speculation, and speculation at different batch sizes can emit different greedy tokens. Identical BF16 logits with a deterministic tie rule give identical tokens, so a divergence always means that the computed logits differ, that their rounding differs, or that ties are handled differently. “A tie” is an observation about where the divergence happened, not an explanation of why. Each recorded divergence is therefore classified by mechanism, with the first kernel at which the computed values differ, and the resulting noise floor is what any comparison between configurations inherits.

[pending: state: the differential test design (configurations, batch shapes, prompts and tokens compared); divergence counts per configuration pair with their mechanism classes; the logprob drift on identical prefixes before a divergence; the effect of `--enable-deterministic-inference` and `--enable-fp32-lm-head`; evidence under `evidence/state/`]

The practical consequence for this paper is that a certified head is compared with the stock head at the same batch shape, never with a stock run at another shape or with an earlier run of a different configuration; otherwise the comparison measures the stock noise floor rather than the certificate.

3.7 Unfinished work is dead, deferred or enclosed

Skipping a computation does not remove its cost unless nothing downstream needs its result. Every skipped result therefore falls into one of three classes. It is *dead* when it has no remaining consumer. It is *deferred* when a later consumer needs it and a reconstruction path exists; the cost is moved, not eliminated, and must be charged where it lands. It is *soundly enclosed* when a certificate bounds it well enough for every consumer that remains. A claimed saving should say which class it belongs to, because early exits and approximations can borrow time from future tokens.

The head is dead work in the relevant sense: once a token is committed, no future computation reads the logits. In target-only decoding of Qwen3.5-4B the same is true of more. The last decoder layer is a full-attention layer (Section 1.1); its attention writes the key–value cache before its MLP runs, and the final MLP, the final norm and the head have no consumer in future state. A certificate that decides the token before that tail could therefore remove about 1.41 GB of the 8.41 GB streamed per step (the head plus the last layer’s MLP, Table 1), and the removal would be permanent. The same cut does not transfer to speculation without re-analysis: the MTP layer and the DFlash drafter consume target hidden states, so work that is dead in plain decoding becomes deferred there. Section 6.7 states the certified-tail hypothesis and its kill test.

3.8 Recurrent state under speculation

Qwen’s hybrid architecture makes state correctness part of verification. Reusable state includes attention KV, recurrent tensors, convolution history, positions and backend metadata. With state orientation $S \in \mathbb{R}^{d_v \times d_k}$, the gated-delta update can be written

$$S_t = \alpha_t S_{t-1} + u_t k_t^\top, \quad u_t = \beta_t (v_t - \alpha_t S_{t-1} k_t), \quad (1)$$

which is the Gated DeltaNet rule $S_t = S_{t-1} \alpha_t (I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$ [89] with the update vector made explicit. The update depends on the preceding state.

Proposition 3.6 (Replay of valid derived records). *If (α_t, u_t, k_t) were recorded along a causal trajectory from S_0 , applying the first a records to that same S_0 reproduces S_a in exact arithmetic:*

$$S_a = \left(\prod_{j=1}^a \alpha_j \right) S_0 + \sum_{t=1}^a \left(\prod_{j=t+1}^a \alpha_j \right) u_t k_t^\top.$$

Proof. Substitute the expansion at $a - 1$ into (1); the new gate multiplies every preceding term and the new outer product adds the last summand. Induction completes the proof. $\square$

The qualification about S_0 matters: replaying a derived u_t from a different state gives the wrong result even when the token IDs agree. The inherited CPU suite contains a one-dimensional witness (a record made at state zero and replayed from state two gives three, where the genuine update gives one). Replaying raw inputs with the true transition is a different operation. Regrouping the recurrence also does not preserve floating-point results; SGLang’s replay fold constrains its operation order and warp count to be a bitwise clone of the verify kernel [65].

A cycle may emit a correction or bonus token before that token has been consumed into every cache, so the runtime must distinguish the *emitted* prefix from the *materialized* state prefix; treating both as one accepted length invites off-by-one state errors. Publication of speculative state should check that the request’s slot and epoch are still current, and memory must not be reused while any stream or captured graph can still write it. An epoch check does not by itself prevent a stale kernel from writing reused memory; ownership and delayed reuse are separate obligations.

[pending: state: differential output-equality tests of speculative against non-speculative decoding at every rejection position, full acceptance, aborts, prefix reuse and slot recycling (H5); evidence under evidence/state/]

3.9 How much recurrent state an exact computation needs

The dense FP32 state is sufficient for exact decoding; nothing yet shows that it is necessary. Two questions should be kept apart: whether less information could be retained, and whether the state has to be materialized as often as it is.

For the first, fix one GDN head of one layer and treat the layer’s inputs as given. Along any permitted history the state evolves as $S_t = S_{t-1}A_t + B_t$ with $A_t = \alpha_t(I - \beta_t k_t k_t^\top)$ and $B_t = \beta_t v_t k_t^\top$, which depend on the inputs and not on the state, and the head reads $o_t = S_t q_t$. A state difference ΔS at time t therefore propagates linearly, to $\Delta S A_{t+1} \cdots A_{t+j}$, and is observed as $\Delta S A_{t+1} \cdots A_{t+j} q_{t+j}$.

Proposition 3.7 (What an exact state must retain). *Let $\mathcal{R}$ be the span of the states reachable over permitted histories and $\mathcal{N}$ the set of differences ΔS with $\Delta S A_{t+1} \cdots A_{t+j} q_{t+j} = 0$ for every permitted continuation and every $j \geq 0$. Two reachable states produce the same outputs under every permitted continuation if and only if their difference lies in $\mathcal{N}$. An exact representation must therefore separate the classes of $\mathcal{R}/(\mathcal{R} \cap \mathcal{N})$, and a linear one needs dimension at least $\dim \mathcal{R} - \dim(\mathcal{R} \cap \mathcal{N})$.*

Proof. Because B_t does not depend on the state, the output difference under a continuation is the linear expression above; it vanishes for every continuation exactly when $\Delta S \in \mathcal{N}$. A representation that merged two states whose difference is not in $\mathcal{N}$ would give the same output where the true outputs differ. A linear map is injective on the quotient only if its rank is at least the quotient’s dimension. $\square$

Treating every input sequence as permitted is conservative, since in the full model a layer’s outputs influence later tokens; it is the notion that makes a reduction safe for any request. Exact witnesses show that neither $\mathcal{R}$ nor $\mathcal{N}$ is favourable without structure in the keys (tests/test_state_structure.py, evidence/state_structure/). Writes with unit coordinate keys and gates inside $(0, 1)$ reach any matrix, so $\mathcal{R}$ can be everything. A direction the current query cannot see is not unobservable: with $\alpha = \beta = \frac{1}{2}$, $k = (\frac{3}{5}, \frac{4}{5})$, $q = e_1$ and $\Delta S = u e_2^\top$ for any value vector u , the observed difference $\Delta S q$ is zero now, but one step later it is $\Delta S A q = -\frac{3}{25}u$. Each transition is a contraction, not an erasure: $\det A_t = \alpha_t^d (1 - \beta_t \|k_t\|^2) > 0$ when $\beta_t \|k_t\|^2 < 1$, so no step forgets a direction exactly. Transitions with non-orthogonal keys do not commute, so there is no common basis in which they are diagonal. And because a gated RMSNorm precedes the output projection, the null space of the projection is not unobservable: two outputs that $W_o = (1, 0)$ cannot tell apart, $(1, 1)$ and $(1, 7)$, project to 1 and $\frac{1}{5}$ after normalization.

One structural reduction is exact. If every key lies in the column space of a fixed orthonormal $U \in \mathbb{R}^{d_k \times r}$ and $S_0 = Z_0 U^\top$, then $S_t = Z_t U^\top$ with

$$Z_t = \alpha_t Z_{t-1} + \beta_t (v_t - \alpha_t Z_{t-1} \kappa_t) \kappa_t^\top, \quad \kappa_t = U^\top k_t,$$

the same recurrence in dimension r , since $U^\top A_t (I - U U^\top) = 0$ and the write $\beta_t v_t k_t^\top$ stays in the form exactly when $k_t \in \text{col}(U)$. The keys are produced by a short convolution, a SiLU and a normalization, and SiLU raises rank (SiLU(log 2 [[1, 2], [2, 4]]) has rank two), so a low-dimensional input does not imply low-dimensional keys; whether real keys are confined to a subspace is a measurement, to be certified by exact or modular rank rather than an SVD threshold [pending: rank and observability audit of captured q, k, α, β traces]. Factored storage $d_v r + r d_k$ beats the dense $d_v d_k$ only below $r = 64$ at $d_v = d_k = 128$, and halves it only at $r \leq 32$ (derived). Floating point adds a further obstruction: an exact rational reduction does not preserve bitwise results, because deferring a decay reassociates FP32 roundings (one ulp apart for $\alpha_1 = \alpha_2 = 0.9$, $x = 1.3$) and rounding an exactly rank-one BF16 outer product to BF16 gives a 32×32 matrix of exact rank 28 (tests/test_state_structure.py, evidence/state_structure/).

The second question has a better answer. A decoder that keeps a dense anchor and the recent actual FP32 operands of each step, and replays them in the reference kernel’s order, reproduces every state word bit for bit by induction, while reading the dense state once per step and writing it once every m steps. That reduces information

Consumer	Required information	Certificate here
Greedy token	Winning index under the reference kernel and tie rule.	Ladder plus gap condition (Sections 3.4, 6.3).
Exact top- k	Winning indices and scores; no partition.	Ladder on membership.
Seeded or fixed-noise sample	Argmax of perturbed scores under the keyed field.	Certified race (Section 7.1).
Acceptance bit	$\exp(z_X)$ against $Uq_X Z$; an interval can suffice.	Guards (Section 7.4; analysis). The engine path uses fixed-noise verification instead (Section 7.3).
Residual sample	The $(p - q)_+$ law, including omitted mass.	Residual races (Section 7.5).
Bonus sample	A sample from the target law under the RNG contract.	Certified race.
Top- p sample	Mass above each token relative to Z .	Partition bounds, or fall back.
Arbitrary log-probabilities	Requested logits and the full partition.	None; ordinary path.

Table 4: Consumers of the head and what each needs. The last rows are where a narrow fast path usually loses.

movement, not information, and it is what replay-based rollback already does [39, 65]. Section 10 treats it as a lever with a pre-registered test.

3.10 Useful progress and elapsed time

Let a cycle commit R output tokens in elapsed time $T > 0$. Under a stationary regenerative model with finite expectations the long-run rate is

$$\lambda = \frac{\mathbb{E}R}{\mathbb{E}T}, \quad (2)$$

the ratio of accumulated reward to accumulated time, not $\mathbb{E}[R/T]$, and not a queueing-latency model. A change improves the rate exactly when $\mathbb{E}R_1/\mathbb{E}R_0 > \mathbb{E}T_1/\mathbb{E}T_0$. Higher acceptance does not compensate for arbitrary overhead, and lower acceptance can be worthwhile when the saved work exceeds the lost progress. The denominator includes drafting, verification, the head, refinement, completion, state commit, non-overlapped dispatch and control. A fixed-width target pass evaluates all admitted positions even when the first proposal is rejected; overlapped kernels require a critical-path measurement rather than a sum of kernel durations.

3.11 What each consumer needs

Table 4 lists the information each consumer of the head requires. A fast path is admitted by these contracts, not by a request being labelled “decode”. Grammar masks, repetition penalties and temperature transforms are part of the scores; they cannot be applied after candidates have been discarded.

4 Where the time goes

A component that occupies a fraction f of a step can speed the step up by at most $1/(1 - f)$, so the head’s share of the time bounds what any head certificate can deliver, and the other shares say where the rest of the stack leaves room. This section measures those shares for SGLang serving Qwen3.5-4B in its production configuration: CUDA graphs for prefill and decode, the overlap scheduler, FlashInfer attention and greedy decoding, at batch sizes from 1 to 128 for plain decoding and from 1 to 32 for MTP speculation.

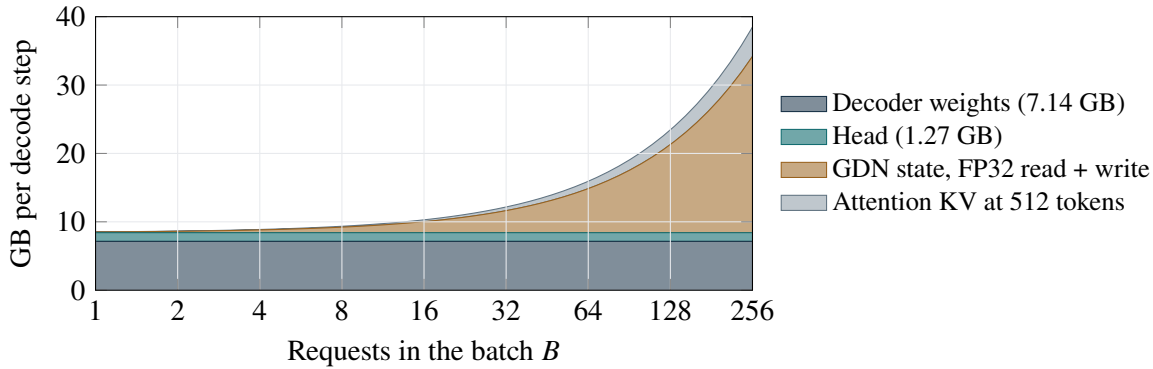


Figure 3: Derived bytes per plain decode step against batch size, from the checkpoint’s tensor sizes and configuration (Table 1): weights and head once per step, 100.7 MB of GDN state and 16.8 MB of attention KV (512 context tokens) per request. A calculation, not a measurement; logits and convolution state are omitted. The measured attribution is [pending: profile, PR #13].

Decode on this GPU is dominated by streaming bytes, so the byte counts of Table 1 give a first picture before any trace. Figure 3 draws the derived bytes of one plain decode step against the batch size at a fixed context. The weights are streamed once whatever the batch; the recurrent state and the attention cache are per request. With FP32 GDN state that the decode kernel reads and writes once per step, the per-request term equals the weights near a batch of 70 at a 512-token context and dominates beyond it. The head’s share of the bytes falls accordingly, from 15% at batch one.

4.1 Attribution method

[pending: profile: workload and profiled windows (fixed concurrency, prefill excluded); nsys session tracing with CUDA-graph node records; step reconstruction from graph-replay correlation IDs; kernel categories and GEMM labelling; idle time split inside and outside replays; profiler perturbation against uncollected twin windows; the measured HBM read peak used as the bandwidth reference; draft in `~/vp-coord/drafts/profile.tex`; evidence under `evidence/profiles/`]

4.2 The head in plain decode and in speculation

[pending: profile: the head chain (GEMM, FP32 copy, argmax) as a share of the plain decode step at $B = 1, 8, 32, 128$; the head microbenchmark of the exact calls SGLang makes at $M = 1$ to 256; the MTP cycle’s target and draft head chains at $B = 1, 8, 32$; evidence `evidence/profiles/attribution/`, `evidence/profiles/head_microbench.json`]

Pending figure. Share of a plain decode step and of an MTP cycle by component (target head chain, draft heads, MTP layer, weight GEMMs, GDN state kernels, attention, small kernels, idle) at each batch size.

Data: `evidence/profiles/breakdown.csv`

4.3 Recurrent state and host gaps

[pending: profile: measured GDN decode-kernel time per request and its implied bandwidth; bytes per step by component with the measured-kernel check; the per-position FP32 state writes of MTP verification; GPU idle time and the host call sites on the critical path of a speculative cycle; evidence `evidence/profiles/bytes_per_step.json`, `evidence/profiles/bytes_per_step_sweep.csv`]

5 Transport on real hidden states

Transport was the central mechanism of the earlier revisions. This section states it with the hypotheses that the proofs need, derives the condition under which it can beat the alternative, and reports what happened when real

draft and target states were compared.

5.1 Tile geometry and already-paid summaries

Let $W \in \mathbb{R}^{V \times D}$ be the head used for both a draft hidden vector h^d and an aligned target hidden vector h^t , both taken at the head input (after the final norm). Write $\Delta = h^t - h^d$, $z_i^d = \langle w_i, h^d \rangle$ and $z_i^t = \langle w_i, h^t \rangle$. A vocabulary partition C covers all admitted tokens with disjoint tiles. For each tile c choose a centre μ_c and a radius

$$r_c \geq \max_{i \in c} \|w_i - \mu_c\|_\infty, \quad a_c = \langle \mu_c, \Delta \rangle, \quad \epsilon_c = r_c \|\Delta\|_1. \quad (3)$$

The draft pass emits $M_c^d = \max_{i \in c} z_i^d$ and $Z_c^d = \sum_{i \in c} \exp(z_i^d)$; stable implementations store a maximum and a scaled mass. The centres and radii are checkpoint metadata; the maxima and masses belong to one request, position and draft vector.

Theorem 5.1 (Transported tile certificates). *In real arithmetic, with the same head matrix for both passes and both masses taken at the same temperature, every $i \in c$ satisfies*

$$z_i^d + a_c - \epsilon_c \leq z_i^t \leq z_i^d + a_c + \epsilon_c. \quad (4)$$

Consequently

$$\max_{i \in c} z_i^t \leq M_c^d + a_c + \epsilon_c =: U_c, \quad (5)$$

$$e^{a_c - \epsilon_c} Z_c^d \leq Z_c^t \leq e^{a_c + \epsilon_c} Z_c^d. \quad (6)$$

Proof. $z_i^t - z_i^d - a_c = \langle w_i - \mu_c, \Delta \rangle$, whose absolute value is at most $r_c \|\Delta\|_1$ by Hölder's inequality. Maximization gives (5); monotonicity of the exponential and summation give (6). $\square$

In the engine the draft maxima and masses come from BF16-rounded draft logits, so (5) and (6) must be widened by the draft kernel's error before they enclose anything the engine computes; Section 6 gives the terms. The Euclidean form replaces r_c by $r_c^{(2)} \geq \max_{i \in c} \|w_i - \mu_c\|_2$ and $\|\Delta\|_1$ by $\|\Delta\|_2$.

5.2 A summary that supports both selection and mass

For a set of labelled logits, keep its top k items under descending score then ascending token index, and the mass of every discarded item. Two such summaries merge by taking the top k of their retained union and adding the newly discarded mass to the two old tails.

Proposition 5.2 (Top- k plus tail is a compositional summary). *For disjoint finite domains the merge produces the same top- k items and tail mass as summarizing the union, and it is associative and commutative in exact arithmetic under the common total order.*

Proof. An item omitted from a local top- k has at least k locally superior items and cannot enter the union's top- k , so every possible winner is retained. The original tails and the newly discarded items partition the final tail, so summing them gives its mass without subtracting nearly equal totals. Every parenthesization and order represents the same union. $\square$

This is a sufficient-statistic construction from familiar reductions, and floating-point tail masses are not associative. Its value is an interface: one epilogue can serve candidate generation, bounds and completion. Retaining a small set R_c per tile also tightens the mass bounds: recompute its target logits exactly and transport only the rest,

$$\sum_{i \in R_c} e^{z_i^t} + e^{a_c - \epsilon_c} Z_{c \setminus R_c}^d \leq Z_c^t \leq \sum_{i \in R_c} e^{z_i^t} + e^{a_c + \epsilon_c} Z_{c \setminus R_c}^d, \quad (7)$$

which is never looser than transporting the whole tile, at the cost of $|R_c|$ extra row projections.

5.3 Geometry decides

A good draft does not imply a small $\|h^t - h^d\|$. The head can ignore directions in which the states differ, and probabilities ignore a common shift of all logits, yet a crude hidden-space norm charges both. With $W = ((1, 0), (0, 0))$ and $\Delta = (0, L)$ both logits are unchanged for every L , while $r_c \|\Delta\|_1$ grows with L ; coordinate radii remove that particular failure. The bounds form a hierarchy: a scalar radius; coordinate radii with $\epsilon_c = \sum_j r_{cj} |\Delta_j|$; grouped norms

$$\epsilon_c = \sum_g r_{cg} \|\Delta_g\|_2, \quad r_{cg} \geq \max_{i \in c} \|(w_i - \mu_c)_g\|_2; \quad (8)$$

and a low-rank split $w_i = \mu_c + Qu_i + e_i$ with the projected part evaluated exactly and a norm bound on e_i . Richer geometry costs metadata and bound arithmetic, and a learned centre or subspace does not certify the residual radius; that radius must cover the stored weights.

5.4 Greedy and top- k admission

Suppose at least k distinct target rows have been evaluated and τ is the k -th best evaluated score. A tile may be skipped only when $U_c < \tau$; then at least k evaluated items outrank every skipped one. The strict inequality preserves unresolved ties. This establishes exact selection; it does not establish an exact unrestricted softmax, because an omitted logit still contributes to the normalizer even when it cannot win.

5.5 What transport must beat

The alternative source of evidence (Section 6) evaluates a low-precision copy $\bar{w}_i = w_i - e_i$ of each row at the target state. In real arithmetic its error for row i is $\langle e_i, h^t \rangle$, bounded by $\|e_i\|_2 \|h^t\|_2$, where e_i is known offline. Transport's error is $\langle w_i - \mu_c, \Delta \rangle$. The two can be compared exactly.

Proposition 5.3 (No better bound from the same information). *For fixed h and radii $\epsilon_b \geq 0$ over a partition of the coordinates into blocks b , $\sup\{|\langle e, h \rangle| : \|e_b\|_2 \leq \epsilon_b \forall b\} = \sum_b \epsilon_b \|h_b\|_2$ and $\sup\{|\langle e, h \rangle| : \|e_b\|_\infty \leq \epsilon_b \forall b\} = \sum_b \epsilon_b \|h_b\|_1$. Likewise $\sup\{\langle w - \mu, \Delta \rangle : \|w - \mu\|_\infty \leq r\} = r \|\Delta\|_1$.*

Proof. The inequalities are Cauchy–Schwarz and Hölder. Equality holds at $e_b = \epsilon_b h_b / \|h_b\|_2$, at $e_b = \epsilon_b \text{sign}(h_b)$ and at $w = \mu + r \text{sign}(\Delta)$. $\square$

Both envelopes are therefore optimal for the metadata they use. For errors not aligned with h the realized $|\langle e_i, h \rangle|$ is typically about $\|e_i\|_2 \|h\|_2 / \sqrt{D}$; closing that gap needs more arithmetic, not a better inequality.

Proposition 5.4 (Exclusion conditions). *After the winner a is scored exactly, let $m_i = z_a - z_i$. Row-level transport excludes i if and only if $m_i > r_c \|\Delta\|_1 - \langle w_i - \mu_c, \Delta \rangle$; tile-level transport excludes tile c if and only if $z_a > M_c^d + \langle \mu_c, \Delta \rangle + r_c \|\Delta\|_1$; self-evidence (in real arithmetic, with envelope β_i) excludes i if and only if $m_i > \beta_i - \langle e_i, h \rangle$. The slack terms lie in $[0, 2r_c \|\Delta\|_1]$ and $[0, 2\beta_i]$.*

Proof. Substitute $z_i^d + \langle \mu_c, \Delta \rangle = z_i - \langle w_i - \mu_c, \Delta \rangle$ and $\tilde{z}_i = z_i - \langle e_i, h \rangle$ into the upper bounds. $\square$

Theorem 5.5 (Transport cannot beat self-evidence unless drift is small). *For every tile c and every centre μ_c , $r_c = \max_{k \in c} \|w_k - \mu_c\|_\infty \geq \frac{1}{2} \text{diam}_\infty(c)$ and $r_c^{(2)} = \max_{k \in c} \|w_k - \mu_c\|_2 \geq \frac{1}{2} \text{diam}_2(c)$.*

1. (Hölder family.) *With b -bit round-to-nearest per-row codes, $s_i = \|w_i\|_\infty / (2^{b-1} - 1)$ and no clipping, $|\langle e_i, h \rangle| \leq \frac{s_i}{2} \|h\|_1$, and $\frac{s_i}{2} \|h\|_1 \leq r_c \|\Delta\|_1$ whenever $\|\Delta\|_1 / \|h\|_1 \geq \|w_i\|_\infty / ((2^{b-1} - 1) \text{diam}_\infty(c))$. If some row of c has the opposite sign, or zero, in the coordinate where $|w_{ij}|$ peaks, then $\text{diam}_\infty(c) \geq \|w_i\|_\infty$ and the condition is implied by $\|\Delta\|_1 / \|h\|_1 \geq 1/(2^{b-1} - 1)$: $1/127$ for int8 and $1/7$ for int4 .*
2. (Euclidean family.) *$\|e_i\|_2 \|h\|_2 \leq r_c^{(2)} \|\Delta\|_2$ if and only if $\|\Delta\|_2 / \|h\|_2 \geq \delta_i^* := \|e_i\|_2 / r_c^{(2)}$.*

Proof. For rows k, l of the tile, $\|w_k - w_l\| \leq \|w_k - \mu_c\| + \|w_l - \mu_c\| \leq 2r_c$. Round-to-nearest codes give $|e_{ij}| \leq s_i/2$, and Hölder gives the first inequality. The rest is algebra, and opposite signs at the peak coordinate give $|w_{ij} - w_{kj}| \geq |w_{ij}|$. $\square$

Proposition 5.6 (Transport against the static screen). *If μ_c lies in the convex hull of the tile’s rows, then $U_c^T - U_c^{\text{static}} \geq r_c(\|\Delta\|_1 - \|h\|_1)$, where $U_c^T = M_c^d + \langle \mu_c, \Delta \rangle + r_c \|\Delta\|_1$ and $U_c^{\text{static}} = \langle \mu_c, h \rangle + r_c \|h\|_1$. Transport is therefore no tighter than the static screen with the same geometry once $\|\Delta\|_1 \geq \|h\|_1$.*

Proof. $U_c^T - U_c^{\text{static}} = (M_c^d - \langle \mu_c, h^d \rangle) + r_c(\|\Delta\|_1 - \|h\|_1)$, and $M_c^d = \max_k \langle w_k, h^d \rangle \geq \langle \mu_c, h^d \rangle$ for a convex combination μ_c . $\square$

The quantity that decides the comparison is the *drift ratio*

$$\rho = \frac{\|h^t - h^d\|_2}{\|h^t\|_2}, \quad (9)$$

measured on the exact head inputs behind the same verified draft token. In the Euclidean family, with 64-row tiles of contiguous token indices and μ_c the tile mean, transport’s envelope for row i is narrower than the per-row int8 self-evidence envelope if and only if $\rho < \delta_i^* = \|e_i\|_2 / r_c^{(2)}(i)$. Table 5 evaluates δ^* on the Qwen3.5-4B head from the weights alone. Its median is 0.85% against per-row int8 and 15.5% against per-row int4. For the manuscript’s original pairing of an ℓ_∞ radius with $\|\Delta\|_1$ (Theorem 5.5, part 1, with the tile radius measured rather than bounded by the diameter), the corresponding threshold on $\|\Delta\|_1 / \|h\|_1$ has a median of 0.30%. The comparison is of envelope widths, not of certification rates, candidate counts or runtime; it uses no hidden states, and it ignores the BF16 rounding of the draft logits that transport also has to carry. The measured drift decides this width comparison, not which mechanism is faster. The Hölder condition of Theorem 5.5 needs $\text{diam}_\infty(c) \geq \|w_i\|_\infty$, which holds for 99.9% of rows with 64-row tiles (evidence/precision/head_constants.json). It says that transport can compete with an int8 copy only if draft and target states agree to better than about one part in a hundred at the head input, which a drafter trained to predict tokens has no particular reason to achieve. The static tile screen is weak on this head for a related reason: the 64-row tile radius exceeds the row norm (median $r_c^{(2)} / \|w_i\|_2 = 1.14$), so $\langle \mu_c, h \rangle + r_c^{(2)} \|h\|_2$ is wider than the plain row bound $\|w_i\|_2 \|h\|_2$, and tiles in token order are barely tighter than random ones. A scalar ℓ_∞ radius paired with $\|\Delta\|_1$ is about four times looser again than the Euclidean pairing for a spread-out Δ (derived from the same constants, $r_c \approx 1.30 \|w_i\|_\infty$ and $\|\Delta\|_1 \approx 0.8\sqrt{D} \|\Delta\|_2$). The enclosures of the two mechanisms can be intersected, so a hybrid is never looser than either; the question is only whether transport’s cheaper evidence excludes enough to pay for itself.

5.6 The measured drift

The decisive measurement captures aligned draft and target head inputs from the running engine, confirms the alignment on traceable cycles (the index that predicts a token is not automatically the index that consumes it), and evaluates ρ and the certification rates of every transport variant on held-out prompts, against the static screen at h^t and against self-evidence on the same states.

[pending: geometry: capture and alignment method; the distribution of ρ for the MTP drafter by draft step, and for the DFlash drafter by block position; certification rates (rows and tiles skipped, greedy and top- k) for scalar, coordinate, grouped and low-rank transport and the static screen; a drift-scaling replay that shrinks Δ to find where transport would start to help; evidence under evidence/head_geometry/]

Pending figure. Empirical distribution of the drift ratio ρ on aligned MTP and DFlash head inputs, drawn against the band of δ^* quantiles from Table 5 (0.67%–1.15% against int8, 12.2%–20.8% against int4); ρ to the left of the band means transport’s envelope is the narrower one.

Data: evidence/head_geometry/rho.csv

Quantity (tied head, $248,320 \times 2,560$)	p10	p50	p90
Row $\ w_i\ _2$	0.574	0.652	0.746
int8 per-row $\ e_i\ _2 / \ w_i\ _2$	0.81%	0.97%	1.25%
int8, 128-column groups	0.64%	0.68%	0.75%
int4 per-row	14.8%	17.5%	22.6%
64-row tile $r_c^{(2)} / \ w_i\ _2$, token-index order	0.99	1.14	1.31
random tiles (control)	1.03	1.19	1.37
δ^* against int8 per-row	0.67%	0.85%	1.15%
δ^* against int8, 128-column groups	0.51%	0.61%	0.72%
δ^* against int4 per-row	12.2%	15.5%	20.8%
Hölder-family threshold on $\ \Delta\ _1 / \ h\ _1$ against int8 per-row	0.23%	0.30%	0.40%

Table 5: Weight-only constants of the Qwen3.5-4B head (float64 on CPU; quantiles over a one-million-row sample; no hidden states). $\delta_i^* = \|e_i\|_2 / r_c^{(2)}(i)$ is the drift ratio ρ below which transport’s Euclidean envelope for row i is narrower than the Euclidean int8 envelope, with 64-row tiles of contiguous token indices and μ_c the tile mean (Theorem 5.5); the last row is the analogous threshold for the ℓ_∞/ℓ_1 pairing, $s_i \|h\|_1 / 2$ against $r_c \|\Delta\|_1$. Widths are compared, not certification rates or runtime. Derived; evidence/precision/head_constants.json.

6 Self-evidence: a certified low-precision head

A cheap approximation of the head can supply the evidence itself. The approximate pass bounds every row; rows whose interval can still matter are re-scored more precisely; exact arithmetic or the stock kernel completes whatever remains. Certified greedy screening of this form exists in public code (Section 2.3). What this section adds is an envelope that covers the floating-point arithmetic of the pass as the GPU performs it, including Hopper tensor-core accumulation under a published model, a refinement procedure that serves greedy selection and sampling alike, and bounds on how much re-scoring each needs.

6.1 Accumulation error

Lemma 6.1 (Accumulation trees). *Let $s = \sum_\ell p_\ell$ be evaluated by a rooted tree whose leaves carry the exact values p_ℓ and whose internal nodes v return $\hat{s}_v$ with $|\hat{s}_v - \sum_c \hat{s}_c| \leq u_v \sum_c |\hat{s}_c| + \eta_v$ over the children c of v . Let d be the largest number of internal nodes on a leaf-to-root path, N the number of internal nodes, $u \geq \max_v u_v$, $\eta \geq \max_v \eta_v$ and $du < 1$. Then*

$$|\hat{s}_{\text{root}} - s| \leq \gamma_d \sum_\ell |p_\ell| + N\eta(1 + \gamma_d), \quad \gamma_d = (1 + u)^d - 1 \leq \frac{du}{1 - du}. \quad (10)$$

A node that multiplies its only child by a constant σ and rounds fits the model after multiplying the leaves below it by σ .

Proof. For a node v and a leaf ℓ below it let $P_{v\ell} = \prod (1 + u_x)$ over the internal nodes x on the path from ℓ up to v , and for an internal node w below v let P'_{vw} be the same product over the nodes strictly above w up to v ($P'_{vv} = 1$). By induction on height,

$$|\hat{s}_v| \leq \sum_\ell |p_\ell| P_{v\ell} + \eta \sum_w P'_{vw}, \quad |\hat{s}_v - s_v| \leq \sum_\ell |p_\ell| (P_{v\ell} - 1) + \eta \sum_w P'_{vw}.$$

Leaves satisfy both with equality. For an internal node, $|\hat{s}_v| \leq (1 + u_v) \sum_c |\hat{s}_c| + \eta$ and $|\hat{s}_v - s_v| \leq \sum_c |\hat{s}_c - s_c| + u_v \sum_c |\hat{s}_c| + \eta$; substituting the hypotheses for the children multiplies every P and P' by $(1 + u_v)$ and adds $P'_{vv} = 1$. Finally $P_{v\ell} \leq (1 + u)^d$, $P'_{vw} \leq (1 + u)^{d-1}$, and there are N internal nodes. $\square$

The IEEE model gives $u_v = u$ for a rounded addition, since $|a+b| \leq |a|+|b|$ [24]. Sequential accumulation from zero has $d = n$, pairwise summation $\lceil \log_2 n \rceil$, blocks of B summed as a tree and then accumulated $\lceil n/B \rceil + \lceil \log_2 B \rceil$, and split- K adds the number of partials. A multi-term adder that aligns $k+1$ addends to the largest, truncates each to F fractional bits below its leading bit, adds exactly and rounds once has $u_v = (k+1)2^{-F} + 2^{1-p}$ with truncation, and depth $\lceil n/k \rceil$. Tensor cores behave this way rather than as IEEE adders [15]. Khattak and Mikaitis model Hopper’s BF16 wgmma with FP32 output as $k = 16$, $F = 25$ with truncation [28], so $u_v = 17 \cdot 2^{-25} + 2^{-23}$ and $\gamma_{D/16} \approx 1.0 \times 10^{-4}$ at $D = 2560$; the conservative default $D \cdot 2^{-22} \cdot 1.001 = 6.1 \times 10^{-4}$ covers it about sixfold. This is a published model, not a documented contract, so kernels built on it are validated against exact arithmetic on adversarial inputs. The same model gives FP8 wgmma only 13 fractional bits in 32-term blocks, consistent with the accumulation limits reported for FP8 training [13]; an FP8 tensor-core pass needs periodic promotion to FP32 and its own analysis. On CUDA cores with IEEE round-to-nearest, γ at $D = 2560$ is about 1.7×10^{-6} for a blocked reduction and plays no role.

6.2 The envelope of the approximate pass

Row i is quantized with group scales $s_{ig} > 0$, integer codes q_{ij} and a set O of columns kept exact. The dequantized row is $\bar{w}_{ij} = s_{ig(j)} q_{ij}$ for $j \notin O$ and $\bar{w}_{ij} = w_{ij}$ for $j \in O$; the error $e_i = w_i - \bar{w}_i$ is computed exactly offline. The pass computes $\hat{t}_g = \sum_{j \in g} q_{ij} h_j$, $\hat{v}_g = \text{fl}(s_{ig} \hat{t}_g)$, sums the groups, adds $\sum_{j \in O} w_{ij} h_j$ and returns $\tilde{z}_i$.

Theorem 6.2 (Envelope). *Suppose every product $q_{ij} h_j$ and $w_{ij} h_j$ is exact in the accumulator format (true for int8 or BF16 weights times BF16 activations in FP32; underflow is covered by η) and every node obeys Lemma 6.1 with (u, η) and depth d . Then $|z_i - \tilde{z}_i| \leq \beta_i$ with*

$$\beta_i = Q_i + \gamma_d \|\bar{w}_i\|_2 \|h\|_2 + N\eta(1 + \gamma_d) \max(1, \max_g s_{ig}), \quad (11)$$

where Q_i may be any of $\|e_i\|_2 \|h\|_2$, $\sum_b \|e_{i,b}\|_2 \|h_b\|_2$, $\|e_i\|_\infty \|h\|_1$ or $\sum_b \|e_{i,b}\|_\infty \|h_b\|_1$ for any partition of the columns into blocks b (in particular O and its complement, so that only the norm of h outside O enters). Round-to-nearest codes without clipping have $\|e_{i,g}\|_\infty \leq s_{ig}/2$.

Proof. $z_i - \tilde{z}_i = \langle e_i, h \rangle + (\langle \bar{w}_i, h \rangle - \tilde{z}_i)$. The first term is bounded blockwise by Cauchy–Schwarz or Hölder. The second is the rounding error of evaluating $\langle \bar{w}_i, h \rangle$ by a tree whose leaves are $s_{ig} q_{ij} h_j$ and $w_{ij} h_j$; apply Lemma 6.1 and $\sum_j |\bar{w}_{ij} h_j| \leq \|\bar{w}_i\|_2 \|h\|_2$. $\square$

The componentwise midpoint-radius form $|z - \tilde{z}| \leq |e|^\top |h|$ is tighter than any of these relaxations and costs the same to evaluate when $|e|$ is stored per group [63]; the norm forms are used here because they need one constant per row or block.

Theorem 6.3 (Integer pass). *Let $h = s_h q_h + e_h$ with s_h a power of two and integer codes, so $s_h q_h$ and e_h are exact. Let $I_i = \sum_j q_{ij} q_{h,j}$ be accumulated exactly in 32-bit integers ($D q_{\max}^w q_{\max}^h < 2^{31}$; $2560 \cdot 127^2$ qualifies) and $\tilde{z}_i = \text{fl}(\text{fl}(s_i \text{fl}(I_i)) s_h)$ without overflow or underflow. Then*

$$|z_i - \tilde{z}_i| \leq Q_i + \|s_i q_i\|_2 \|e_h\|_2 + \gamma_2 \|s_i q_i\|_2 \|s_h q_h\|_2.$$

Proof. $z_i = s_i s_h I_i + s_i \langle q_i, e_h \rangle + \langle e_i, h \rangle$. The two roundings give $|\tilde{z}_i - s_i s_h I_i| \leq \gamma_2 s_i s_h |I_i|$, and the last multiply by a power of two is exact. Apply Cauchy–Schwarz to I_i and to $\langle q_i, e_h \rangle$. $\square$

This path needs no model of floating-point accumulation at all, because int8 products accumulated in int32 are exact for this shape [1]; it pays instead for quantizing the activation.

Lemma 6.4 (Computed norms). *In FP32 with round to nearest, let $\hat{\sigma}$ be any computed sum of $\text{fl}(h_j^2)$ with depth $d \leq 2^{14}$, $\hat{r} \geq (1 - 2^{-16})\sqrt{\hat{\sigma}}$ any square root, and $N_2 = \text{fl}(\text{fl}(\hat{r}(1 + 2^{-10})) + 2^{-50})$. If $D \leq 2^{14}$ then $N_2 \geq \|h\|_2$. The same holds for $N_1 = \text{fl}(\text{fl}(\hat{\sigma}_1(1 + 2^{-10})) + 2^{-50}) \geq \|h\|_1$ with $\hat{\sigma}_1$ a computed sum of $|h_j|$.*

Proof. Nonnegative additions return at least $(1 - u)$ times their exact sum, less μ under flush-to-zero, so $\|h\|_2^2 \leq (1 - u)^{-d} \hat{\sigma} + 3D\mu$ and $\|h\|_2 \leq (1 - u)^{-d/2} (1 - 2^{-16})^{-1} \hat{r} + 2^{-54}$ for $D \leq 2^{14}$. The computed $N_2 \geq (1 - u)^2 (1 + 2^{-10}) \hat{r} + 2^{-51}$, and $(1 - u)^2 (1 + 2^{-10}) \geq (1 - u)^{-d/2} (1 - 2^{-16})^{-1}$ because $du/2 \leq 2^{-11}$. $\square$

Lemma 6.5 (Outward rounding of the computed interval). *In a format with $p \geq 12$ and round to nearest, let $\tilde{z}$ be representable, let β^* be a real bound with $|z - \tilde{z}| \leq \beta^*$, and let b be computed from representable upper bounds by at most $k \leq 59$ rounded operations $(+, \times, \sqrt{})$ on nonnegative values along any path of the expression, so that $b \geq (1 - u)^k \beta^* - k\mu$. Set*

$$\hat{b} = \text{fl}(\text{fl}(\text{fl}(b(1 + 2^{6-p})) + \text{fl}(2^{4-p}|\tilde{z}|)) + 2^{e_{\min}+26}), \quad lo = \text{fl}(\tilde{z} - \hat{b}), \quad hi = \text{fl}(\tilde{z} + \hat{b}).$$

Then $lo \leq z \leq hi$, so every comparison between computed endpoints is sound.

Proof. Each rounding of a nonnegative value loses at most a factor $(1 - u)$ and μ , so $\hat{b}(1 - u) \geq (1 - u)^4 (1 + 64u)b + 16u(1 - u)^3 |\tilde{z}| + (1 - u)^2 2^{e_{\min}+26} - 4\mu$. Also $lo \leq \tilde{z} - \hat{b}(1 - u) + u|\tilde{z}| + \mu$. It suffices that $\hat{b}(1 - u) \geq \beta^* + u|\tilde{z}| + \mu$: the $|\tilde{z}|$ terms satisfy $16u(1 - u)^3 \geq u$, the b terms satisfy $(1 - u)^{4+k} (1 + 64u) \geq (1 - 63u)(1 + 64u) \geq 1$ for $u \leq 2^{-12}$, and $2^{26}\mu$ absorbs the absolute terms. The upper endpoint is symmetric. $\square$

For FP32 the constants are $1 + 2^{-18}$, $2^{-20}|\tilde{z}|$ and 2^{-100} . Offline constants ($\|e_i\|_2$, $\|\bar{w}_i\|_2$, $\|w_i\|_2$) are computed exactly and stored rounded up. The exact CPU reference emulates every BF16, FP32 and FP64 rounding on rationals and checks each bound against real arithmetic, including witnesses that the accumulation term, the outward rounding and the norm inflation are each necessary (tests/test_precision.py, evidence/precision/).

6.3 Certified argmax and its cost

Theorem 6.6 (Refinement ladder). *Let levels $\ell = 0, \dots, L$ supply enclosures $lo_i^\ell \leq z_i \leq hi_i^\ell$, exact at level L , and let f_i be nondecreasing maps. Starting from a row set S , at each level let $\tau = \max_{i \in S} f_i(lo_i^\ell)$ and keep $\{i \in S : f_i(hi_i^\ell) \geq \tau\}$; stop when one row remains, when every kept interval is a point, or at level L , and return the smallest kept index. The result is $\min \arg \max_{i \in S} f_i(z_i)$. With $f_i = \text{id}$ this is R-real, with $f_i = \text{rn}_B$ it is R-bf16, and with $f_i(x) = x/T + g_i$ it is S-race. The token does not depend on the precision or reduction order of any level.*

Proof. $[f_i(lo_i), f_i(hi_i)]$ encloses $f_i(z_i)$. A maximizer m is never dropped: for the row j that attains τ , $f_m(hi_m) \geq f_m(z_m) \geq f_j(z_j) \geq f_j(lo_j) = \tau$. A dropped row is strictly below the row attaining τ . If one row remains it is the maximizer. If all kept intervals are points, each kept value is at least τ and at most τ , so all kept rows are maximizers, and every maximizer is kept. The exact level produces points. The returned index is a function of the maximizer set alone. $\square$

The ladder decides R-real (or S-race); Theorem 3.5 then converts an R-real decision with enough margin into the stock kernel's decision, and a row without that margin runs the stock kernel (Figure 2). The Lean formalization proves the certificate, its uniqueness across levels and both filter properties in scaled integers (formal/CertifiedArgmax.lean).

Proposition 6.7 (Candidate counts). *Let the level-0 intervals have widths $\omega_i = hi_i - lo_i \leq \omega$ (with outward rounding, $\omega_i \approx 2\beta_i$) and let C be the rows kept at level 0.*

1. *Greedy:* $C \subseteq \{i : z_i \geq z_{\max} - \omega_i - \omega_a\}$ for any maximizer a , hence $|C| \leq e^{2\omega/T'} / p_{\max}^{(T')}$ for every $T' > 0$, where $p^{(T')} = \text{softmax}(z/T')$.
2. *Race:* if the g_i are i.i.d. standard Gumbel and $p = \text{softmax}(z/T)$, then with $t = 2\omega/T$,

$$\mathbb{E}|C| \leq \sum_i \frac{p_i e^t}{1 + p_i(e^t - 1)} \leq \min(V, e^t).$$

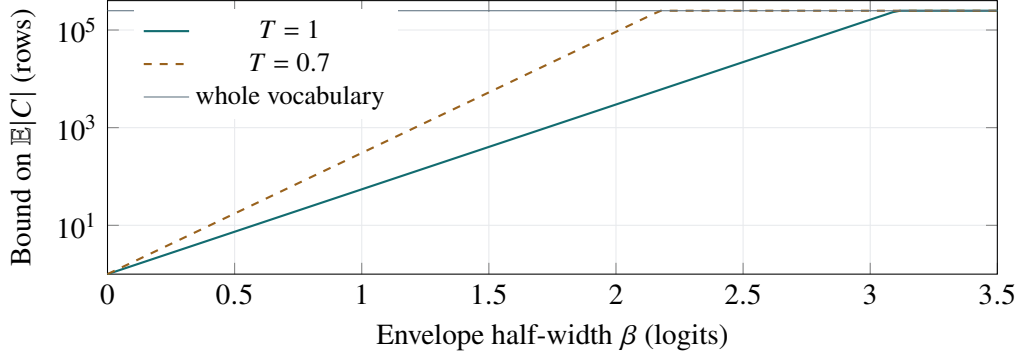


Figure 4: The distribution-free bound $\min(V, e^{4\beta/T})$ on the expected number of rows an exact Gumbel race must re-score (Proposition 6.7, with $\omega \approx 2\beta$). An analytic bound, not a measurement; measured candidate counts on real states are [pending: geometry].

Proof. (1) $hi_i \leq z_i + \omega_i$ and $lo_a \geq z_a - \omega_a$, so a kept row satisfies $z_i + \omega_i \geq \tau \geq z_a - \omega_a$. Each such row has $p_i^{(T')} \geq p_{\max}^{(T')} e^{-2\omega/T'}$, and the probabilities sum to at most one. (2) In score space the widths are ω_i/T , so $C \subseteq \{i : y_i \geq M - t\}$ with $y_i = z_i/T + g_i$ and $M = \max_j y_j$. Row i belongs to that set if and only if $y_i + t \geq \max_{j \neq i} y_j$. The maximum of independent Gumbels with locations z_j/T is Gumbel with location $\log \sum_{j \neq i} e^{z_j/T}$, and two independent Gumbels with locations a, b satisfy $P(G_a \geq G_b) = e^a / (e^a + e^b)$. Hence the probability is $p_i e^t / (p_i e^t + 1 - p_i)$; sum over i . $\square$

At $T = 1$ and envelope half-width β (so $t \approx 4\beta$), the expected re-scoring set for exact sampling is at most 55 rows for $\beta = 1$, 2,981 for $\beta = 2$ and up to 1.6×10^5 for $\beta = 3$ (Figure 4). Exact sampling through the cheap pass is therefore cheap only if the envelope is about two logits wide or less. Greedy selection has no distribution-free bound: a flat distribution (small $p_{\max}$) needs many rows. The envelope in logits is $\beta_i \approx \|e_i\|_2 \|h\|_2$; with the int8 constants of Table 5 that is about $0.0063 \|h\|_2$, so the norm of the head input at the served positions decides whether β is a fraction of a logit or several logits [pending: geometry: distribution of $\|h\|_2$ at the head input and of β per variant].

6.4 Candidate sets on real hidden states

[pending: geometry: candidate-set sizes per head variant (int8 per row, int8 in 128- and 32-column groups, FP8 E4M3, int4) and envelope family on held-out plain-decode and MTP head inputs; distinct candidate rows in real co-scheduled batches; Gumbel-race candidates at $T = 1$ and $T = 0.7$; the share of positions whose re-scoring still overlaps under each accumulation model; outlier-exact columns, rotations and a centred head as ablations; evidence under evidence/head_geometry/]

6.5 Sparse rows do not imply sparse weight traffic

For rows i of a microbatch let A_i be the set of retained vocabulary rows or tiles. A kernel that reuses weights across rows reads the union $\bigcup_i A_i$, not the average of the masks. If each row independently retained a tile with probability f , the batch would need that tile with probability

$$1 - (1 - f)^B. \quad (12)$$

At $f = 1/16$ and $B = 32$ this is about 87.3%, so a per-row skip fraction of 93.75% can coexist with nearly a full weight stream. Self-evidence is less exposed to this than transport, because its saving is in the bytes of the cheap pass, which every row shares, and its candidates are few rows rather than whole tiles; the union matters for the re-scoring gather. Figure 5 is an analytic model, not a measurement of mask correlation.

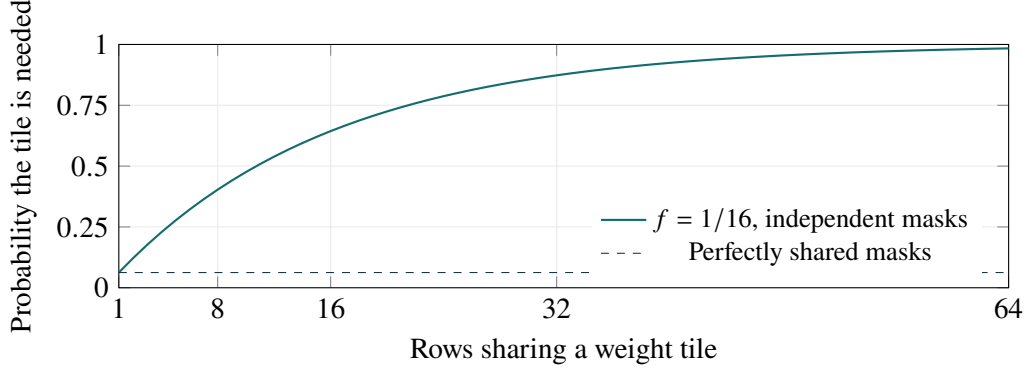


Figure 5: A model of batch-union traffic, not a benchmark. Correlation among masks decides whether per-row sparsity becomes reduced weight traffic.

6.6 Kernel and head-path runtime

A certified head pays off only if the whole path beats the strongest dense head chain at the served shapes:

$$T_{\text{cheap pass}} + T_{\text{bounds}} + T_{\text{compaction}} + T_{\text{re-score}} + \Pr[\text{fallback}] T_{\text{stock head}} < T_{\text{stock head chain}}. \quad (13)$$

The right-hand side is the stock GEMM plus the FP32 copy and the argmax or top-1 that follow it; a fused dense epilogue that removes the copy and the argmax is a comparator in its own right, as is FlashSampling [62]. Equation (13) is a measurement criterion, not a prediction. If the head chain is a fraction f of a step, removing it entirely gives at most $1/(1-f)$ overall under otherwise unchanged execution.

[pending: kernel: the int8 layout and scales, the bound epilogue, candidate compaction, BF16 re-scoring and its accumulation contract, the graph-safe conditional fallback, and measured head-path time against the stock chain at $M = 1$ to 256 with repeats and variance; draft in `~/vp-coord/drafts/kernel.tex`; data evidence/certified_head/head_path_time.csv]

Pending figure. Head-path time against the number of rows M : stock cuBLAS BF16 GEMM plus FP32 copy plus argmax, the certified head (measured and expected), and the int8 pass alone, with p10–p90 bands.

Data: evidence/certified_head/head_path_time.csv

6.7 Beyond the head: a certified tail (hypothesis)

Section 3.7 observed that in target-only decoding the last layer’s MLP, the final norm and the head have no consumer in future state. The hypothesis, from a later proposal, is that a cheap surrogate of that tail with a sound enclosure could let the commit rule decide the token before the tail runs, removing up to 1.41 GB of the 8.41 GB streamed per step (derived, Table 1). A one-shot surrogate that costs C (surrogate plus certificate) and certifies a fraction p of positions pays only if $C < p T_{\text{tail}}$, and the ceiling for a tail fraction f of the step is $1/(1-f)$. The directional margin (Proposition 3.2) is the natural certificate when the surrogate predicts the head input with a radius.

The kill test is deliberately optimistic. Record, on about 2,000 positions from 16 prompts of mixed domains, the residual stream at the cut, the reference head input, the logits and the tokens; freeze an int8 tail surrogate; take the oracle radius $\|h - \tilde{h}\|_2$; and apply the directional certificate with no overhead. If the oracle certification rate p_{oracle} satisfies $p_{\text{oracle}} T_{\text{tail}} \leq C_{\text{surrogate}}$, the one-shot design cannot pay and is dropped; otherwise real bounds are compared against the dense tail and against the certified head alone, which is the decisive ablation. The cut does not transfer to speculation without re-analysis, because the MTP layer and the DFlash drafter consume the target’s hidden states. [pending: kill test for the certified tail; evidence path to be assigned]

7 Exact sampling and speculative decisions

Greedy selection is the easy consumer. This section turns the same evidence into sampled decisions: a token drawn from the target law, an accept or reject bit for a drafted token, and the residual or bonus sample that follows. The evidence may come from transport or from self-evidence; every statement below takes enclosures of the logits or of their exponentials as input and does not care which mechanism produced them.

7.1 A certified race under a keyed noise field

Theorem 7.1 (Certified race). *For any fixed noise vector g and $T > 0$, the ladder of Theorem 6.6 with $f_i(x) = x/T + g_i$ returns the S -race token, which is the token of a dense race with the same g and tie rule. If the g_i are i.i.d. standard Gumbel and independent of z , the token has law $\text{softmax}(z/T)$.*

Proof. Theorem 6.6 and the Gumbel-max identity [21, 54]; ties have probability zero under continuous noise. $\square$

Using bounds to avoid evaluating most perturbed scores is the principle of A* Sampling [43] and of lazy Gumbel sampling from a retrieved top set [45]. What the fixed field adds is pathwise equality with the dense race, which a lazily instantiated field does not give. Top- k sampling applies the ladder to membership (drop i when h_i is below the k -th largest l_o) and races over the certified set; equality then also needs the engine’s tie semantics at the k -th position. Min- p admits i when $z_i \geq z_{\max} + T \log(\min-p)$, which intervals decide without a partition function. Top- p admission depends on the mass above each token relative to Z , needs partition bounds, and is usually cheaper to hand to the ordinary path.

Proposition 7.2 (RNG contract). *(i) Seeded equality between a certified and a dense procedure holds when both evaluate the same function of the same noise values with the same tie rule. This requires a counter-based field, a pure function of (seed, request, absolute position, draw purpose, token) that the certified path can evaluate for any token in any order; a stream consumed in evaluation order does not qualify, and in the exact reference assigning a stream to rows in candidate order changed the winner in 247 of 300 cases (evidence/precision/precision_tests.json). (ii) Equality in distribution needs i.i.d. noise with the correct marginal, independent of the proposal draw, the acceptance uniform and the scores; a residual race also needs independence from the acceptance event. (iii) Fixed-noise verification (Section 7.3) needs the noise of a position to be the same whether the position is drafted or decoded plainly, so it must be keyed by absolute position, not by cycle or batch slot. (iv) A hash with 2^{32} levels makes the law exact only for the idealized continuous field; the stock seeded sampler shares that approximation.*

The stock seeded sampler (Section 3.2) already has the form (i) requires. FlashSampling’s noise offset depends on the tiling [62], so it cannot serve as a certified path’s field without change.

7.2 How much of a sample can be emitted before the scores are known

A race needs the winner’s interval to separate. A different construction can emit a token with some probability before any candidate is separated, and then finish the rest of the law after more computation. Let $\mathcal{P}(I)$ be the set of target laws compatible with the information I held so far.

Proposition 7.3 (Common mass). *Consider procedures that, before acquiring more information, emit token i with probability r_i (the same for every $p \in \mathcal{P}(I)$) and otherwise continue until p is known. Such a procedure outputs exactly p for every $p \in \mathcal{P}(I)$ if and only if $r_i \leq \inf_{p \in \mathcal{P}(I)} p_i$ for every i , and then it may finish by sampling the residual $(p - r)/(1 - \alpha)$ with $\alpha = \sum_i r_i$. The largest immediate probability is*

$$\alpha^*(I) = \sum_i \inf_{p \in \mathcal{P}(I)} p_i.$$

Proof. If the procedure is exact, the final probability of i is at least the probability r_i of emitting it immediately, for every compatible p ; so $r_i \leq \inf_p p_i$. Conversely, if $r_i \leq \inf_p p_i$ then $p - r \geq 0$ and sums to $1 - \alpha$ for every compatible p , so the residual is a law, and the output law is $r + (1 - \alpha) \cdot (p - r)/(1 - \alpha) = p$. $\square$

For two candidate laws $\mathcal{P} = \{p, q\}$, $\alpha^* = \sum_i \min(p_i, q_i) = 1 - \text{TV}(p, q)$, the overlap of a maximal coupling, which is exactly the acceptance probability of speculative sampling [7, 34]. The proposition says the same idea applies to any set of laws the evidence leaves open, and that nothing can do better without more information.

A two-token example shows that exact sampling can finish where greedy selection cannot. Suppose the evidence only says that the probability of token 1 lies in $[0.49, 0.51]$. Then $\alpha^* = 0.49 + 0.49 = 0.98$. Draw U uniform on $[0, 1]$: emit token 1 if $U < 0.49$, token 0 if $U \geq 0.51$, and otherwise compute p and emit token 1 exactly when $U < p$, reusing the same U . The output has law p , and 98% of draws never need the scores; the greedy decision, by contrast, is undecided because the interval contains $1/2$. For two tokens this procedure is inverse-CDF sampling with bounds on the CDF. In general it gives equality in distribution, not seeded equality with a dense race.

Proposition 7.4 (Bounded logit error). *Let $p = \text{softmax}(z/T)$ and $q = \text{softmax}(\tilde{z}/T)$, and suppose $\max_i(z_i - \tilde{z}_i) - \min_i(z_i - \tilde{z}_i) \leq \Delta$. Then $p_i \geq e^{-\Delta/T} q_i$ for every i . Consequently, with $\alpha = e^{-\Delta/T}$:*

1. *sampling q with probability α , and otherwise computing p and sampling $(p - \alpha q)/(1 - \alpha)$, samples p exactly;*
2. *always sampling q gives $\text{TV}(p, q) \leq 1 - \alpha$.*

Proof. Write $d_i = z_i - \tilde{z}_i$ with $m \leq d_i \leq M$ and $M - m \leq \Delta$. Then $p_i/q_i = e^{d_i/T} / \sum_j q_j e^{d_j/T} \geq e^{m/T} / e^{M/T} \geq e^{-\Delta/T}$. Part 1 is Proposition 7.3 with $r = \alpha q$. For part 2, $\text{TV}(p, q) = \sum_i (q_i - p_i)_+ \leq \sum_i (1 - \alpha) q_i = 1 - \alpha$. $\square$

With self-evidence the range is at most twice the largest envelope half-width, so $\alpha \geq e^{-2 \max_i \beta_i/T}$. The trade-off against the race is different: this sampler needs the cheap pass's full softmax and, with probability $1 - \alpha$, the whole exact head, whereas the race re-scores a candidate set whose expected size grows like $e^{4\beta/T}$ and returns the seeded token.

Proposition 7.5 (Approximate mode: KL and sequence total variation). *Suppose $\min_c \|z - \tilde{z} - c\mathbf{1}\|_\infty \leq \varepsilon$ at temperature one (at temperature T replace ε by ε/T). Then $\text{KL}(p \| q) \leq \varepsilon^2/2$ and $\text{KL}(q \| p) \leq \varepsilon^2/2$, so $\text{TV}(p, q) \leq \varepsilon/2$. If an approximate decoder samples q_t in place of p_t at every position and the bound holds with ε_t at every reachable prefix, the laws P and Q of the first N tokens satisfy $\text{KL}(P \| Q) \leq \frac{1}{2} \sum_t \varepsilon_t^2$ and*

$$\text{TV}(P, Q) \leq \frac{1}{2} \left(\sum_{t=1}^N \varepsilon_t^2 \right)^{1/2}.$$

Proof. Let $d = z - \tilde{z} - c\mathbf{1}$ with $\|d\|_\infty \leq \varepsilon$; the shift c changes neither law. Then p is q tilted by d : $p_i = q_i e^{d_i} / \mathbb{E}_q[e^d]$. Let $\psi(\lambda) = \log \mathbb{E}_q[e^{\lambda d}]$, so $\psi(0) = 0$ and $\text{KL}(p \| q) = \mathbb{E}_p[d] - \psi(1) = \psi'(1) - \psi(1) = \int_0^1 \lambda \psi''(\lambda) d\lambda$. Here $\psi''(\lambda)$ is the variance of d under the tilted law $q^{(\lambda)}$, at most $(2\varepsilon)^2/4 = \varepsilon^2$ because d takes values in an interval of length 2ε . Hence $\text{KL}(p \| q) \leq \varepsilon^2/2$; the reverse divergence follows by exchanging the roles of p and q . Pinsker's inequality gives $\text{TV} \leq \sqrt{\text{KL}/2} \leq \varepsilon/2$. For sequences, the chain rule gives $\text{KL}(P \| Q) = \sum_t \mathbb{E}_P[\text{KL}(p_t \| q_t)] \leq \sum_t \varepsilon_t^2/2$, and Pinsker's inequality again gives the total-variation bound. $\square$

To guarantee a sequence total variation of at most 0.01 over $N = 1,024$ tokens with a uniform bound, the proposition needs a logit error, up to a common shift, of at most $\varepsilon = 6.25 \times 10^{-4}$, far tighter than any int8 envelope in Section 6. An approximate head is therefore a quality trade-off to be measured (Section 10); these bounds cannot certify it as harmless over long outputs.

7.3 Fixed-noise verification

Theorem 7.6 (Fixed-noise verification). *Let $G_{t,i}$ be keyed by (seed, absolute position t , token i) and i.i.d. standard Gumbel across (t, i) . Define $Y_t = \min \arg \max_i (z_{t,i}(Y_{<t})/T + G_{t,i})$. A speculative decoder that accepts a drafted token at position t if and only if it equals $\arg \max_i (z_{t,i}(x_{<t})/T + G_{t,i})$ at the drafted prefix, and emits that argmax at the first mismatch, outputs Y for every drafter and every depth policy, and Y has the target’s autoregressive law.*

Proof. By induction on positions, every accepted token equals the argmax at a prefix equal to $Y_{<t}$, so it is Y_t ; at the first mismatch the prefix is $Y_{<t}$ and the emitted token is Y_t . The drafter only changes how many positions one target pass verifies. Given $Y_{<t}$, $G_{t,\cdot}$ is independent of $z_t(Y_{<t})$, so $Y_t \mid Y_{<t} \sim \text{softmax}(z_t(Y_{<t})/T)$. $\square$

This is the drafter-invariant construction of Daliri, Musco and Suresh [12]. For the certified head it matters because verification becomes a perturbed argmax, which Theorem 7.1 certifies with no partition function and no residual sampler. SGLang’s seeded sampler already keys noise by (seed, position, token), so a fixed-noise verifier would reproduce seeded plain decoding token for token, up to the stock rounding of log softmax at near-ties.

Proposition 7.7 (Acceptance under fixed-noise coupling). *If the drafter proposes $\arg \max_i (\log q_i + G_i)$ and the target accepts if and only if it equals $\arg \max_i (\log p_i + G_i)$, then*

$$P(\text{accept}) = \sum_i \left(\sum_j \max\left(\frac{p_j}{p_i}, \frac{q_j}{q_i}\right) \right)^{-1}, \quad \frac{1}{2} \sum_i \min(p_i, q_i) \leq \sum_i \frac{p_i q_i}{p_i + q_i} \leq P(\text{accept}) \leq \sum_i \min(p_i, q_i).$$

Proof. Both choose i if and only if $G_i > \max_{j \neq i} (G_j + m_j)$ with $e^{m_j} = \max(p_j/p_i, q_j/q_i)$, an event of probability $(1 + \sum_{j \neq i} e^{m_j})^{-1}$. The bounds follow from $\max(x, y) \leq x + y$, from $\max(x, y) \geq x$ and $\geq y$ summed over j , and from $p + q \leq 2 \max(p, q)$. $\square$

Per position, fixed-noise acceptance is at most the $\min(1, p/q)$ rate $1 - \text{TV}(p, q)$ and at least half of it; the two agree when $p = q$. Whether the cheaper certificate pays for any lost acceptance is an empirical question [pending: acceptance under fixed-noise coupling against standard verification on real MTP and DFlash drafts at $T = 1$ and $T = 0.7$].

The sampling contract adopted here. The certified speculative path uses fixed-noise verification with exactly the stock seeded sampler’s noise: $G_{t,i} = -\log(-\log x_{t,i})$ in FP64, with $x_{t,i}$ the 32-bit MurmurHash of (seed, absolute position t , token i) scaled to $[0, 1]$, as in `multinomial_with_seed` (Section 3.2). A drafted token is accepted if and only if it equals the target’s keyed sample at its position, and at the first mismatch that sample is emitted. Three consequences follow. For SGLang’s default point-mass drafts the per-position acceptance probability is unchanged: the stock rule accepts a draft d when a uniform coin falls below $p(d)$, and the keyed rule accepts it when d wins the race, which also has probability $p(d)$. With a seed, the output is token-identical to stock seeded decoding without speculation, computed from the stock kernel’s logits (Theorem 7.6 under R-stock), provided that every row whose race the certificate cannot decide under the stock rounding of log softmax runs the stock sampler; stock speculative sampling is equal to non-speculative decoding only in law, because its coins and its inverse-CDF residual consume randomness differently. Without a seed the output equals stock in law. Top- p requests fall back to the stock path. The acceptance guards below are analysis and are not built: under the stock-token contract the stock inverse-CDF residual would still need a dense head row per verify step. The design is decided; its measurement is [pending: kernel: token-for-token agreement with stock seeded decoding on replayed steps, and acceptance against the stock speculative sampler].

7.4 Acceptance guards from incomplete mass

For one reached proposal $X = x$ write $w_x = e^{z_x}$ and $Z = \sum_i e^{z_i}$, and draw $U \in [0, 1)$ with the reference convention. For $q_x > 0$, acceptance is equivalent to

$$U q_x Z < w_x, \tag{14}$$

and the $\min(1, \cdot)$ need not be evaluated because $U < 1$ already handles $p_x \geq q_x$.

Theorem 7.8 (Acceptance guards with enclosures). *Let $w_x \in [N^-, N^+]$ and $Z \in [Z^-, Z^+]$ with $N^- > 0$, let $0 < q_x \leq 1$ be the actual proposal mass and $U \in [0, 1)$. Then $Uq_x Z^+ < N^-$ implies acceptance, $Uq_x Z^- \geq N^+$ implies rejection, the two never both hold, and replacing an enclosure by a tighter one cannot reverse a certified decision. For U uniform and independent of the enclosures, neither guard fires with probability*

$$P_\gamma = \min(1, N^+/(q_x Z^-)) - \min(1, N^-/(q_x Z^+)).$$

If $N^- = N^+ = w_x$ and every row outside a re-scored set R has a mass enclosure whose upper and lower ends differ by a factor at most κ , then with $\pi = \sum_{i \notin R} p_i$,

$$P_\gamma \leq \frac{p_x}{q_x} \cdot \frac{(\kappa - 1)\pi}{1 - \pi(1 - 1/\kappa)}.$$

Proof. The implications are transitivity with (14); both firing would give $N^+ \leq Uq_x Z^- \leq Uq_x Z^+ < N^-$. A tighter enclosure stays on the same side of the true values. The unresolved uniforms form $[N^-/(q_x Z^+), N^+/(q_x Z^-)) \cap [0, 1)$. For the bound, $P_\gamma \leq (w_x/q_x)(Z^+ - Z^-)/(Z^+ Z^-)$; $Z^+ - Z^- \leq \sum_{i \notin R} e^{z_i}(\kappa - 1)$ because each lower mass is at most e^{z_i} ; and $Z^- \geq Z - \sum_{i \notin R} e^{z_i}(1 - 1/\kappa)$ because each lower mass is at least e^{z_i}/κ . Divide by Z . $\square$

These guards are analysis in this revision: the engine path adopted above uses fixed-noise verification, which needs no partition function, and a certified version of the stock rule would still pay for a dense residual row. Transported tile masses (Theorem 5.1) and exponentiated self-evidence intervals (Theorem 6.2) both supply $Z^\pm$. With self-evidence $\kappa = e^{2\hat{\beta}}$ times the outward rounding of the exponential; at $\beta = 1$ ($\kappa \approx 7.4$), keeping P_γ below 10% at $p_x/q_x \approx 1$ requires the unre-scored mass π to be below about 1.5%. After an adaptive refinement chosen using U , the probability statement no longer applies, although the guards remain sound pointwise. The Lean formalization proves the guards with enclosed numerators and their exclusivity (`formal/CertifiedArgmax.lean`).

7.5 Residual and bonus work

A rejected proposal still requires a sample from $(p - q)_+$, and knowing that rejection occurred does not supply one. When all drafted tokens are accepted, a bonus token needs a target sample. A certificate that decides acceptance cheaply and then completes the whole head for the residual saves little; Section 7.6 shows this happening. Two facts about the engine and one construction reduce the problem.

First, SGLang’s default sampled verification uses point-mass proposals (Section 3.2). With q a point mass at x , the residual is the target restricted to the untried tokens, a race over $V \setminus \{x\}$ that needs no partition function. Second, under fixed-noise verification there is no residual at all. Third, for a proposal with small support there is an exact bounded race.

Let $Q = \{i : q_i > 0\}$. With unnormalized target weights w_i and partition Z , the residual can be sampled from weights

$$r_i = (w_i - Zq_i)_+, \quad \frac{r_i}{\sum_j r_j} = \frac{(p_i - q_i)_+}{\sum_j (p_j - q_j)_+}, \quad (15)$$

and $r_i = w_i$ outside Q . Given $Z^- \leq Z \leq Z^+$, positive-part monotonicity gives $(w_i - Z^+ q_i)_+ \leq r_i \leq (w_i - Z^- q_i)_+$ inside Q .

Theorem 7.9 (Bounded residual race). *Assume valid enclosures of the target weights and of Z , the actual proposal law, a nonzero residual, and a priority field $\pi_i = e^{G_i} > 0$ that is fixed and independent of the proposal draw, the acceptance uniform and the rejection event. A refinement procedure that returns only an interval-certified ordered maximizer of $r_i \pi_i$, or completes the dense race, returns the same token as the dense race. With i.i.d. standard Gumbel G_i it samples the residual law (15) exactly.*

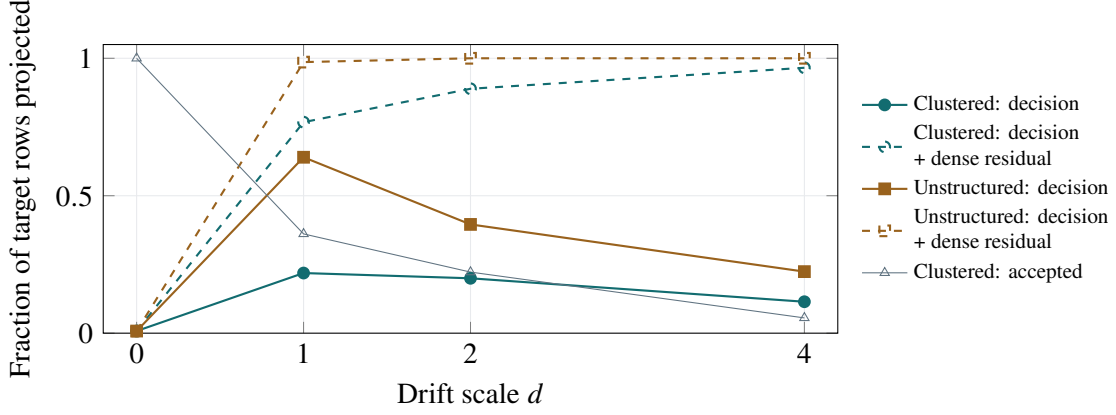


Figure 6: Work counts on constructed heads, not timings (data/synthetic_drift.csv, evidence/synthetic_drift.json). Solid lines count the rows needed to decide acceptance; dashed lines add a dense residual completion after each rejection. The thin line is the fraction of proposals accepted in the clustered family. Evidence production, bounds, bonus sampling and the backbone are excluded.

Proof. The enclosures of r_i multiplied by $\pi_i > 0$ enclose the priorities. A certified maximizer dominates every possible competitor, including the actual one, and a completed evaluation yields the dense ordered maximizer. The distributional statement is Gumbel-max applied to the nonzero residual weights. $\square$

Theorem 7.10 (Residual races with self-evidence). *Replace the enclosures above by $r_i \in [(N_i^- - Z^+ q_i)_+, (N_i^+ - Z^- q_i)_+]$ for $q_i > 0$ and $r_i \in [N_i^-, N_i^+]$ for $q_i = 0$. The conclusions of Theorem 7.9 hold. For a point-mass proposal at x , $r_x = 0$ as soon as $Z^- \geq N_x^+$, and the residual is a race over $V \setminus \{x\}$ that needs no partition bound. SGLang’s target-only rule (accept sibling j if $UZ < \sum_{l \leq j} w_{x_l}$; otherwise sample the target restricted to untried tokens) is certified by the guards with a cumulative numerator and an exclusion race.*

Proof. Positive-part monotonicity gives the enclosures; the rest is the proof of Theorem 7.9. For the point mass, $w_x - Zq_x = w_x - Z \leq 0$. $\square$

The engine samples that residual by inverse CDF, so a race matches it only in distribution. For a support chosen after the draft pass, retaining the top $K + 1$ perturbed entries per tile tightens the outside-support maximum after excluding up to K support tokens; it is not needed for validity, since $r_i \leq w_i$ makes the plain tile maximum an upper bound. For a bonus sample the construction is simpler still: a certified race over the target scores with the bonus position’s field, with no partition function.

7.6 A cheap decision is not a cheap verification

The inherited CPU experiment makes the last point concrete. It constructs two families of 128×8 integer heads with 16-row tiles and 36 paired fixtures each: one clusters rows around separated centres, the control uses unstructured rows. For each fixture $h^t = h^d + d v$ with drift scale $d \in \{0, 1, 2, 4\}$; proposals are sampled from the anchor head’s law, and all probability arithmetic is exact rational. The experiment counts distinct target rows projected, which is work, not time.

At zero drift the summaries describe the target exactly and the acceptance decision needs only its candidate row, an identity check rather than a forecast. At nonzero drift the unstructured head defeats greedy screening. More subtly, large drift makes rejection easy to certify, which lowers the rows needed for the decision while destroying acceptance itself, and completing the residual then erases the saving (Figure 6). At drift four the clustered family decides with 11.4% of the rows but needs 96.5% once the dense residual is charged. The experiment does not refute transport on trained heads; it refutes the inference that a cheap acceptance decision means cheap verification. The residual races above are the answer it motivates, and they remain to be measured.

7.7 A demand frontier over a fixed protocol

For a chain of L proposals each head row is accepted, rejected or unresolved. The first certified rejection at position r proves that no row after r can affect the cycle’s output, while earlier unresolved rows still matter because one of them may be the true first rejection.

Proposition 7.11 (Safe suppression after a certified rejection). *If every issued certificate equals the decision of a fixed reference verifier on its row, the first true rejection is no later than the first certified rejection. Suppressing head work strictly after the latter does not change the reference output, provided completion and state commit use the actual first rejection.*

Proof. At the certified rejection the reference decision is rejection; an earlier unresolved row may also reject, and no later row can be the first rejection. All accepted outputs, the replacement token and the state prefix are determined by rows up to the first rejection. $\square$

The statement is for chains. SGLang’s trees share one coin across siblings and choose the next coin by the accepted node, so a tree frontier must follow the tree’s paths. The target backbone has already run over every admitted position, so the frontier saves head work, not transformer layers.

7.8 When adaptive verification changes the sampling law

Avoiding work whose result is already determined differs from choosing which sampled proposals count. DSpark states that lossless speculation requires admission decisions that do not depend on future candidate tokens, and D-cut gives a counterexample and a repair [11, 41]. A two-token illustration: let $p = q = (1/2, 1/2)$, draw $X \sim q$, discard it and draw afresh from p if $X = 0$, and admit it to the ordinary verifier if $X = 1$. Since $p = q$ an admitted proposal is always accepted, and the output law is

$$\Pr(Y = 0) = \frac{1}{4}, \quad \Pr(Y = 1) = \frac{3}{4}. \quad (16)$$

The acceptance formula was unchanged; admission selected proposals by their realized value. A mandatory progress floor on the first token does not help, since the same construction applies at the second position. Three settings avoid the issue: greedy decoding; target-only verification with point-mass proposals, where there is no proposal law to bias; and fixed-noise verification (Theorem 7.6), under which every depth policy yields the same output. The demand frontier above changes only the amount and order of arithmetic used to decide a fixed protocol, so it is exact pointwise.

7.9 Measurements

[pending: geometry, kernel: Gumbel-race candidate counts through the int8 pass on real states at $T = 1$ and $T = 0.7$; mass-enclosure ratios and P_7 for standard acceptance on real MTP and DFlash drafts; the fraction of decisions resolved before any re-scoring]

8 Drafting

The verifier’s work per committed token falls as the drafter’s accepted length rises, so the drafter is an axis of the stack in its own right, and it interacts with the head: every draft step of the MTP layer and every DFlash block projects through the same 1.27 GB head.

The native MTP layer is one decoder layer (0.241 GB) that consumes the target’s last hidden state and the embedding of the latest token and then runs the full head. SGLang serves it through the EAGLE worker with a chain of draft steps and one draft token per step in the configuration used here. The public DFlash drafter for this target [8, 92] (revision 9a1996cc) has six Qwen3-style layers of width 2,560, five with sliding-window attention and one with full attention, drafts blocks of up to 16 tokens in one pass, and conditions on the target’s hidden

states after layers 1, 5, 9, 13, 17, 21, 25 and 29. It has no head of its own; SGLang projects its drafts through the target’s tied head. Its model card reports, on one B200 in SGLang with greedy decoding and thinking enabled, 3.40–4.60× over autoregressive decoding at concurrency 1 with blocks of 16, and 2.15–2.61× at concurrency 32 with blocks of 8, against at most 2.31× and 1.80× for the MTP layer [92]. These are the authors’ numbers on other hardware.

[pending: drafter: DFlash on GH200 at block sizes 4, 8 and 16: acceptance probability by block position and domain, accepted length against concurrency, and the split of a cycle into draft forward, verification and host time; evidence under evidence/drafter/]

[pending: drafter: whether a targeted fine-tune is justified by a measured limitation (longer accepted blocks, cheaper drafting, or draft states that make certificates tighter), judged by total serving time]

Pending figure. Acceptance probability by position within a DFlash block, per block size and domain.

Data: evidence/drafter/acceptance_by_position.csv

9 Exact repair of long windows

A drafter’s window is verified once and then discarded after the first mismatch. Two later proposals ask whether the target’s work on a long window can be reused instead. Both keep exactness by the same device: only tokens that an exact audit pass of the target has verified are committed, approximate states never enter the committed cache, and the noise of a position is never resampled. Both are hypotheses under test in this revision, and their kill tests are designed so that a negative answer is cheap to obtain.

Long-window repair (P2). Greedy continuation is a triangular system $y_i = F_i(y_{<i})$. A parallel repair step $y_i^{(r+1)} = F_i(y_{<i}^{(r)})$ from a window initialized by the drafter commits the verified prefix plus the target’s token at the first mismatch; suffix outputs are new guesses, not certified outputs or reusable states. After r sweeps at least r positions are correct, so repair converges within the window, but nothing guarantees one or two sweeps suffice. This is Jacobi decoding applied to a drafted window; lookahead and consistency-trained decoders pursue related fixed-point iterations [17, 31]. For sampling, fixed per-position random inputs held across repairs give the target law. The gated-delta recurrence composes associatively, so a verifier could reconstruct state only at the accepted boundary instead of materializing every position; whether SGLang already does so is part of the test. An end-to-end speedup S with a non-decode fraction f needs a decode speedup s with $1/(f + (1 - f)/s) = S$: for $S = 5$, $s = 5.44$ at $f = 2\%$ and $s = 9$ at $f = 10\%$ (derived).

Target-anchored residual decoding (P3). For each expensive linear operator $z = Wx$, retain an anchor $(\bar{x}, \bar{z} = W\bar{x})$ from a full target pass over a candidate block and a fixed orthonormal basis U_W with WU_W precomputed; after a few token corrections, repair with $\tilde{z} = \bar{z} + (WU_W)U_W^\top(x - \bar{x})$, leaving out $W(I - U_WU_W^\top)(x - \bar{x})$. Fixed bases matter: a basis computed from the current correction would stream W again. Nonlinearities are evaluated exactly on the repaired activations, and an exact audit pass commits only verified tokens, so exactness holds whatever the quality of the repair. The open question is whether one changed token leaves most downstream activations close enough to the anchor for the audit to accept long prefixes while paying for both full passes, the repairs and the activation cache.

What the repair experiment must show. Local low rank does not imply useful repair of a whole model. Three things must hold separately: the fixed cheap operators must stay accurate after candidate tokens change, not only near the anchor; their errors must stay below the decision margins after attention, nonlinearities, the recurrent state and later layers; and each sweep must resolve enough positions to pay for itself and for the audit. A sufficient condition links these quantities. Let m_i be the exact top-two logit margin at position i , ε_i a bound

Axis	Levers	Status
Progress per verify pass	MTP and DFlash drafting; deeper chains; trees where the GDN path allows; n-gram or suffix drafting; relaxed acceptance (lossy)	[pending]
Bytes per verify pass	Certified int8 head (exact); FP8 W8A8 or W4A16 target (lossy); FP8 KV (lossy); lower-precision or replayed GDN state (lossy unless shown exact)	[pending]
Bytes per draft step	Near-free draft head (hot vocabulary, low-rank or low-precision draft head; lossless end to end because the target verifies); lower-precision draft layer	[pending]
Concurrency and batching	State memory and the 133-request cap; graph batch sizes; chunked prefill; speculation depth by batch size	[pending]
System overheads	Host gaps and synchronizations; fusion; attention backend per shape; overlap	[pending]

Table 6: Axes of the serving stack and the levers tested on each. A lever is exact when it keeps the target’s decisions under the contracts of Section 3; a lossy lever changes them and is measured against a declared quality budget.

on the logit error the approximate evaluator makes at i when its inputs are exact, b_j a bound on the error in the current guess’s effect at an earlier position j ($b_j = 0$ once j is correct), and C_{ij} the influence of position j on the logits at i . If $2(\varepsilon_i + \sum_{j<i} C_{ij}b_j) < m_i$, position i is correct after the next sweep, by the triangle inequality. Two consequences follow. A persistent approximation error η in a recurrence with contraction factor $a < 1$ leaves an error floor of about $\eta/(1-a)$, and a faithful evaluator can advance only as fast as exact Jacobi iteration, which on a copy chain $F_i(y) = y_{i-1}$ gains one position per sweep even with exact rank-one operators. Expected accepted length is best read through hazards: with h_i the conditional failure rate at position i given an accepted prefix, $\mathbb{E}A = \sum_a \prod_{i \leq a} (1 - h_i)$, so the experiment reports prefix advance per sweep and the ratio $R_i = 2 \|\tilde{z}_i - z_i\|_\infty / m_i$ ($R_i < 1$ guarantees the same decision) on two cascading replays against the original anchor, not hidden-state error. The economic gate is $\mu_R/C_R > \mu_0/C_0$ in committed tokens per unit cost, and a lower bound on repair cost with $C_R^{\text{lower}}/(B+1) \geq C_0/\mu_0$ rejects the design outright. The decisive control is a small target-conditioned drafter given the same anchor information and matched in compute, memory traffic, training data, verifier width and target calls; if it does as well, the supported claim is better target-conditioned drafting, not cheap repair of a trajectory. This is analysis that fixes the measurements, not a result.

[pending: repair: Stage A oracle for P2 and P3 with perfect continuation blocks at widths 16 to 128 and batch 32 to 256, timing target verification plus the state commit needed to continue; Stage B with real DFlash proposals and corrections, a dev/held-out split, bases fitted on dev only, and controls (full-target Jacobi from the same initialization, a comparable compact drafter, the initializer without repair); exact token equality after every commit; draft in `~/vp-coord/drafts/repair.tex`]

10 Stacking every axis

The certified head and a better drafter each attack one term of the verifier’s cost. When the head is a sixth of a plain step’s bytes and the recurrent state dominates at high concurrency, a paper about avoided work also has to ask what the whole stack can gain when every term is attacked, which of those gains keep the target’s decisions, and what the others cost in quality. Table 6 lists the axes and levers in scope. Each lever is built as a separate flag or arm so that it composes, and compositions are measured in the baseline, A , B , $A+B$ pattern rather than by multiplying isolated speedups: a lossy target lowers acceptance, a cheaper verifier makes drafting relatively more expensive, and quality losses from separate approximations can compound.

Two frontiers. The *exact stack* uses only techniques that keep the target’s decisions: the certified head, draft-side approximations (the target still verifies), better drafting, and capacity and system work. The *lossy stack* uses everything, under a quality budget declared before the measurement: at most 1.0 point of GSM8K accuracy lost, at least 98% teacher-forced top-1 agreement with the BF16 reference, and at most 0.01 nats of mean top-20 KL

divergence from it. The quality cost of the combined lossy stack is measured directly, not summed from its parts. Proposition 7.5 explains why the budget is empirical: a per-token logit error small enough to certify a long output as close in total variation is far below what any low-precision lever delivers.

Writing the recurrent state less often. Rounding-preserving live replay (Section 3.9) is exact and cuts dense-state traffic per step from $2D$ (read and write) to $D + D/m$ when the state is flushed every m steps, 0.625 of the original at $m = 4$, for a few percent more storage (derived). If the recurrent kernel is a fraction f of a decode step, a kernel twice as fast gives at most $1/(1 - f/2)$, removing it entirely $1/(1 - f)$, and the $m = 4$ traffic cut alone $1/(1 - 0.375f)$ (Amdahl). From bytes alone, f would be about 0.55 at batch 128 and a 512-token context (12.9 GB of FP32 state against 8.41 GB of weights and 2.1 GB of attention KV; derived, Table 1), which would give 1.38×, 2.2× and 1.26×; the measured time share is [pending: profile: GDN kernel share of a batch-128 step]. The pre-registered test decodes 512 tokens greedily at batch 128 from 2,048-token prompts with FP32 state and no speculation, first compares every state word and GDN output of the replayed and dense paths (one differing word rejects bit-exactness), and then times both: the claim is at least 1.10× end-to-end decode throughput, rejected if the upper end of the paired 95% confidence interval falls below 1.10×. SGLang’s own replay path qualifies only if it passes the bit-exactness check. [pending: moonshot: bit-exactness check and paired timing]

Sharing computation across requests. Factoring hidden or recurrent state across unrelated requests does not survive analysis. An exact factorization with a dense residual costs the full projection, so it saves only if $r(d + B) + z < Bd$ with a zero or sparse residual; SiLU and gating raise rank; full-rank pairwise state differences force private factors at least as large as the dense state; one gated-delta step takes four identical states to batch rank four (tests/test_state_structure.py, evidence/state_structure/); online factorization costs of order B^2 times the update; and reassociation breaks equality with the reference. The one exact candidate is structural: the first GDN layer’s input projections (12,352 outputs) depend only on the current token, so they could be read from a precomputed per-token table (5.71 GiB for the whole vocabulary) with convolution and recurrence kept private. Its ceiling is about 0.75% of the dense projection work per token (derived), and at batch 128 even a 99% table hit rate leaves about 72% of batches with a miss. The pre-registered criterion rejects it unless the stage’s measured share of a decode step allows a 1% end-to-end gain [pending: profile: measured share of the layer-0 input projections, PR #13].

[pending: moonshot: derived ceilings per lever stack from bytes and FLOPs per token and the measured HBM peak; measured decode-step times to $B = 1024$ per state and weight format; the lever table with speed and quality; the interaction matrix; draft in ~/vp-coord/drafts/moonshot.tex; evidence under evidence/moonshot/]

Pending figure. Bytes and step time per token against batch size: derived floors for each lever stack, with measured step times as points.

Data: evidence/moonshot/ceilings.csv

11 Serving results

The decisive test of every mechanism in this paper is the served latency–throughput frontier against the strongest optimized baseline, with outputs checked. This section fixes the protocol; the results are pending.

11.1 Protocol

Each server arm is launched once at its fixed maximum capacity, with CUDA graphs and the overlap scheduler confirmed on, and client concurrency is swept against it. The baselines are plain decoding, the native MTP configuration after tuning of its steps and draft tokens, and the public DFlash drafter at its best block size; the main denominator for any speculative change is the strongest of these, not plain decoding. A diagnostic run

that forces synchronization is not a baseline. Prompts come from a frozen mixed workload with disjoint tuning, confirmation and warm-up splits, the prefix cache is flushed between points, and fixed-length throughput runs are kept separate from natural-stopping quality runs.

For request i let n_i be the server-reported number of output tokens and e_i the interval from sending the request to its last non-empty streamed chunk. The frontier plots

$$x = \frac{1}{N} \sum_i \frac{n_i}{e_i}, \quad y = \frac{\sum_i n_i}{t_{\text{last completion}} - t_{\text{first send}}}, \quad (17)$$

so x is the mean per-request end-to-end output rate, which includes the time to first token, and y is the output throughput of the one GPU. A decode-only rate that excludes the first token and a steady-state throughput that excludes the drain are reported alongside. One token per streamed chunk is never assumed. AIPerf is the load generator [47]. Time to first token, inter-token latency, accepted length, failures and CUDA-graph coverage are kept for every point, and failures stay in the report.

Each confirmatory point is repeated with alternating arm order on an otherwise idle machine, and the variation across repeats is reported; CPU-heavy work elsewhere on the host is detected and recorded, because the scheduler and the client are single-threaded Python loops that share the CPU. Exact changes are checked for output equality against the stock configuration at the same batch shape (Section 3.6); lossy changes are measured against the quality budget of Section 10.

11.2 Frontiers

[pending: bench: tuned MTP and DFlash configurations; the baseline frontiers with repeat variance; the quality baseline; draft in `~/vp-coord/drafts/bench.tex`; evidence `evidence/bench/confirm/frontier.csv`]

[pending: bench, integration: before/after frontiers for the certified head in MTP drafting, verification and plain decoding, with output-equality checks (H4)]

[pending: bench, moonshot: the exact-stack and lossy-stack frontiers with their ablations; evidence `evidence/moonshot/frontier_exact.csv`, `evidence/moonshot/frontier_lossy.csv`]

Pending figure. Tokens per second per user against tokens per second per GPU across client concurrencies for plain decoding, tuned MTP, DFlash, the exact stack and the lossy stack, with repeat ranges.

Data: `evidence/bench/confirm/frontier.csv`

12 What did not work, and the limits of what did

12.1 What the review of the earlier revision overturned

Several claims of the earlier revision did not survive review, and they are recorded here rather than silently removed.

- The counterexample showing that sampled verification depth chosen from realized proposals biases the output law was presented as a contribution. DSpark and D-cut state the condition with their own counterexamples [11, 41]; it is now an illustration (Section 7.8).
- “Progressive precision” was proposed without citation. Certified greedy screening with a low-precision head and exact re-scoring is implemented publicly [6, 22, 33, 69], and the bound-then-refine pattern is classical [35, 83]. The claim is narrowed accordingly (Section 2.8).
- The static tile screen proposed as an ablation is CSV-Decode’s mechanism [40], and on this head the tile radius exceeds the row norm, so the screen is looser than a per-row bound (Table 5).
- Statements about the engine were out of date at the pin: plain and seeded sampling are already races, seeded noise is counter-keyed, and default sampled verification uses point-mass drafts, whose residual needs no

partition function (Section 3.2).

- “Exact” was defined against real arithmetic. It is now defined against the stock kernel at the same batch shape, and the evidence that a certified token equals the stock token is conditional on a modelled error bound for that kernel (Section 3.4).
- The Lean file could not be compiled when the earlier revision was written. Lean 4.19.0 now elaborates it and a second file for the certified decisions (`formal/STATUS.md`, `evidence/precision/lean_check.log`).

12.2 Failure criteria

Table 7 pairs each hypothesis with the fastest test that could damage it and the response if it fails. A failed hypothesis stays in the paper with its mechanism; the programme stays ambitious by testing stronger alternatives, not by weakening the meaning of correctness or changing the comparison after the fact.

Hypothesis	Fastest damaging test	Response if it fails	Status
Transport is tighter than self-evidence and the static screen	Aligned real-head replay: ρ against δ^* (Section 5.5)	Drop transport as the head’s evidence source; keep it only if an intersected hybrid pays for its metadata	Crossover derived from weights; ρ [pending]
Certificates save target work	Count rows, re-scoring and completion, not only decision cost	Reject acceptance-only speed claims; measure residual races separately	Cheap decision $\neq$ cheap verification on constructed heads (Section 7.6) [pending]
Certified tokens equal stock tokens	Same-shape token equality on captured states; measured cuBLAS error against the modelled G_i	Widen G_i or send more rows to the stock fallback	[pending]
Fewer bytes read means less time	Head-path time against the stock chain at each served shape	Keep the dense head at shapes where it wins	[pending]
Per-row sparsity reduces weight traffic	Batch unions of candidates and measured HBM traffic	Group compatible rows or stay dense	[pending]
The head is a large share of the step (H1)	nsys attribution	Move effort to the dominant term (state, host gaps)	Bytes derived; time [pending]
Speculative state is equivalent (H5)	Every rejection position, aborts, prefix reuse	Correct the state path before timing	[pending]
Adaptive sampled depth stays exact	Exhaustive tiny laws with candidate-dependent stopping	Non-anticipating policies or fixed-noise verification	Settled by prior art and Theorem 7.6
Recurrent state compresses exactly	Exact witnesses; rank audit of captured keys	Reduce state writes instead of state size	Exact witnesses (<code>evidence/state_structure /</code>); audit [pending]

Hypothesis	Fastest damaging test	Response if it fails	Status
Sharing across requests pays	Measured share of the one structural candidate	Reject	Rejected in analysis; share [pending]
Results transfer	New prompts, a second model or GPU	Narrow the claim and explain the dependence	[pending]

Table 7: Failure criteria and their current status.

12.3 Scope

The measurements concern one model, one engine commit and one GPU, and the experiments are text-only on a multimodal checkpoint. The agreement of certified tokens with the stock kernel rests on a published model of Hopper tensor-core accumulation and on the absence of reduced-precision split- K reductions in the kernels the engine selects; both are tested, neither is a vendor guarantee. The exact CPU references test the certificates on synthetic heads with small vocabularies and widths; they are tests, not proofs, and the Lean files cover the decision logic and scaled-integer bounds, not IEEE rounding, exponentials or probability. The head constants of Table 5 use no hidden states, so they decide nothing about certification rates on their own. Every end-to-end result is pending in this revision.

12.4 What the completed paper should establish

The question was what work a verifier actually needs. The mathematics now says precisely what evidence suffices for each consumer, how to state exactness against the kernel an engine actually runs, and when each of the two evidence sources can be tighter than the other. The measurements will say which of them survives real hidden states, real kernels and real batching on this model, and what the rest of the serving stack contributes when every axis is pushed. A useful outcome includes a well-explained negative one: transport failing on real geometry, or a certified head that reads half the bytes yet loses to the stock chain at the shapes that matter, would each rule out a tempting argument with a mechanism attached.

A Engine integration notes

The source audit uses SGLang commit `bd66ce343e4f6e2f2b75d7e820fe4d0718a8d824`, a frozen research pin rather than the latest revision. The observations below come from reading that code; the integration is proposed engineering work unless a later section reports it measured.

Path under <code>python/sglang/</code>	Observation and proposed integration
<code>srt/layers/logits_processor.py</code>	<code>_compute_lm_head</code> is the single hook through which plain decoding, the MTP drafter and the verifier compute logits: <code>torch.matmul</code> with BF16 output by default, <code>torch.mm(..., out_dtype=float32)</code> under <code>--enable-fp32-lm-head</code> . A certified head replaces this call for admitted consumers only; log-probability requests and unsupported transforms keep the ordinary path.
<code>srt/models/qwen3_5_text.py</code>	The Qwen3.5 head is tied to <code>embed_tokens</code> , which is not a <code>ParallelLMHead</code> , so SGLang’s quantized-head paths do not apply; a low-precision copy must be separate and the BF16 head stays resident for re-scoring and fallback.

Path under python/sglang/	Observation and proposed integration
srt/models/qwen3_5_mtp.py	set_embed_and_head skips the head when embeddings are tied, so a reduced draft head from --speculative-token-map does not reach the MTP drafter while indices are still remapped [pending: moonshot: test and patch].
srt/layers/sampler.py	Unseeded sampling is an exponential race; seeded sampling is a Gumbel race keyed by (seed, position, token) in FP64 (multinomial_with_seed). The certified race uses the same field.
srt/models/dflash.py	DFlash projects draft hidden states through the target head for its candidates; this is the draft-side summary producer for transport.
kernels/ops/attention/fla/gdn_replayssm_spec_fold.py	The replay fold is written as a bitwise clone of the verify kernel and constrains operation order and warp count; its accept_lens include the bonus token in that backend's indexing [65].

B Mechanisms carried over but not tested in this revision

The earlier revision developed a broader programme around the head. Nobody is testing these parts in this revision, so they are condensed here with their proved statements; they are hypotheses, not results.

The DFlash 2 selector. At the inspected implementation adjacent candidate scores have the form

$$S_t(a, b) = U_t(b) + \sum_{r=1}^{d_s} A(a)_r H(h_t)_r B(b)_r, \quad (18)$$

with a gate that depends on the draft hidden state [26, 67]. A greedy walk needs only the realized row $S_t(x_{t-1}, \cdot)$, which reduces score work from $O(BLK^2 d_s)$ to $O(BLK d_s)$ at the price of a dependent sequence of small contractions. Alternatively the lattice can emit, for every predecessor, its winning successor $f_t(a) = \arg \max_b S_t(a, b)$, after which the path is $x_t = (f_t \circ \dots \circ f_1)(x_0)$.

Proposition B.1 (Finite-map execution). *Given the same row winners, serial map walking and prefix composition produce the same path. A work-efficient scan of length L over K -entry maps uses $O(LK)$ map work and $O(\log L)$ composition stages, excluding the score producer.*

Proof. The t -th prefix composition applies exactly the first t transitions to the anchor, and function composition is associative. A work-efficient tree scan uses $O(L)$ compositions of $O(K)$ indexed evaluations each, with logarithmic depth. $\square$

Function scanning is standard; the question would be its profitable placement. The inherited CPU reference implements the simpler $O(LK \log L)$ Hillis–Steele scan and checks path equivalence only.

Prefix value. On a fixed lattice with a substochastic surrogate $r_t(a, b)$ for the probability of matching the next target token, $J_r(x) = \sum_{j=1}^L \prod_{t \leq j} r_t(x_{t-1}, x_t)$ is maximized by $V_t(a) = \max_b r_t(a, b)(1 + V_{t+1}(b))$ with $V_{L+1} = 0$. If the estimate $\hat{r}$ also lies in $[0, 1]$ and edge errors are uniformly bounded by ϵ_t , telescoping gives $|J_r - J_{\hat{r}}| \leq \delta = \sum_t (L - t + 1) \epsilon_t$ and optimizer regret at most 2δ . LiLiCorr's finding that an exact best-path search over a different score decodes worse than the greedy rule [64] is the relevant warning.

A compiler interface for evidence and effects. The earlier revision proposed typed interfaces (head summaries, decision demands, state leases, evidence with an enclosure contract) so that illegal compositions become visible: a top- k summary cannot satisfy a residual-sampling demand, a lease cannot be reused after cancellation, and a real-arithmetic enclosure cannot discharge a bitwise requirement without a numerical bridge. Search over implementations [85] would operate inside that boundary. No part of it is implemented.

Scheduling and structural tuning. If predicted progress $F_i(d)$ is known and cost depends only on the total optional depth m , the snapshot problem is solved by

$$G_i(m) = \max_{0 \leq d \leq L_i} \{G_{i-1}(m-d) + F_i(d)\}, \quad \max_m \frac{G_B(m)}{C(m; z)}, \quad (19)$$

which is exact only under that cost hypothesis. Graph tiers make cost discontinuous: with rewards (1, 1.9, 2.7) and costs (1, 2, 2) at depths (0, 1, 2), depth one looks worse than zero while depth two is best, so stopping at the first local decline fails. Autotuning a kernel that mutates state must restore that state between trials [76].

GPU execution families. A certificate path can run as a two-stage mask-then-compact pipeline, as a bounded number of guarded refinement rounds inside a CUDA graph, or as a persistent work queue; ordinary inter-block spin barriers are not a legal substitute for global synchronization. Graph-safe designs need fixed buffers, device-resident masks and stable addresses, and a guarded stage that does nothing still costs a launch [77]. Nsight Systems establishes the critical path and Nsight Compute the traffic of selected kernels, whose replay and cache control change what is measured [50]. Hopper’s asynchronous warp-group MMA requires sm_90a, and Blackwell’s data-centre and consumer targets expose different instruction sets and limits [46, 48, 51]; implementation backends range from Triton to TileLang and CuTe [52, 75, 76].

C Mathematical extensions and small falsifiers

Transformed logits. If a transform adds a known per-token correction b_i with tile bounds $b_c^- \leq b_i \leq b_c^+$, then $z_i^d + a_c - \epsilon_c + b_c^- \leq z_i^t + b_i \leq z_i^d + a_c + \epsilon_c + b_c^+$, and the mass bounds multiply by $e^{b_c^+}$. A mask that deletes unknown members cannot preserve the unmasked lower mass; a safe lower bound may collapse to zero.

One anchor mass cannot follow a temperature change. The unnormalized masses (1, 3) and (2, 2) both sum to four, while their squares sum to ten and eight, so a mass at a new temperature cannot be reconstructed from the old sum alone.

A conditional rate certificate. If positive intervals enclose expected progress and time for a candidate and a baseline, $R_1^-/T_1^+ > R_0^+/T_0^-$ (with $T_0^- > 0$) certifies $\lambda_1 > \lambda_0$ in (2). Benchmark confidence intervals are not deterministic enclosures, so the inequality exposes what must be estimated rather than certifying a speedup.

Agreement at an anchor proves nothing away from it. Let $h^d = (1, 0)$ and $h^t = (1, 1)$. Rows with the same first coordinate have identical anchor logits while their second coordinates make the target logits arbitrarily different, so an anchor maximum or mass alone cannot certify the target; a geometric bound, more projection or exact completion is necessary.

RNG tests. Exhaustive tiny-law tests should cover point-mass proposals, zero support, $p = q$, acceptance at the comparison boundary, all-accept cycles, first-position rejection and candidate-dependent stopping, with proposal draws, acceptance uniforms and residual noise aligned across the compared procedures.

D CPU evidence carried over from the earlier revision

The earlier revision’s exact references use bounded integer head weights and hidden vectors, with unnormalized mass 2^z represented exactly in rational arithmetic; this is an exponential model with logits scaled by $\log 2$ and avoids trusting floating exponentials. Eighteen test methods in the decision reference and three in the race reference pass, with four local tests of the benchmark client; Table 9 lists the generated finite instances inside them. The counts are not independent confirmations of anything about a trained model, and the race tests use fixed positive rational priorities, so they establish pathwise equivalence of the bounded and dense algorithms, not continuous Gumbel randomness. Seeds, commands and results are in `evidence/decision_tests.json`, `evidence/race_tests.json` and their logs.

Check	Executed count	Scope
Top- k plus tail composition	250	Exact union and associative merge.
Transported tile bounds	300 heads; 1,048 tiles	3,787 weight rows enclosed.
Greedy certified screening	250	Same winner under a deterministic total order.
Retained-tail tightening	565 tiles	Exact mass enclosed; bounds no looser.
Acceptance interval guards	1,600	Certified bits agree with complete mass.
Unresolved interval measure	160	Exact agreement with Theorem 7.8.
Lazy acceptance	500	With and without retained seeds; finite completion.
Coupled one-step sampling	200	Same output with exact residual fallback.
Rejection-correction identity	300 pairs	Includes zero-probability support entries.
Demand frontier	1,092 partial states	5,460 complete decision assignments.
Exclusion-aware race summaries	300	Exact maxima after support exclusions.
Target priority race	250	Same token as a dense priority race.
Sparse-support residual race	250	Exact bounded and dense race equivalence.
Finite-map execution	400	Scan, walk and composition agreement.
Gated-delta replay	150 trajectories	868 accepted-prefix replay checks.
Lease reuse	300 schedules	Sequential epoch and base-prefix guards.

Table 9: CPU evidence inherited from the earlier revision, on `main`. Counterexamples for biased depth selection, graph-cost cliffs, batch unions, invalid numerical bounds, support truncation and stale namespaces are also included. No GPU implementation is exercised.

The theory workstream adds an exact reference that emulates BF16, FP32 and FP64 rounding on rationals and checks every bound and decision of Sections 6 and 7 against real arithmetic: 20 test methods and 39,761 generated checks, including 5,600 accumulation-bound checks over eight summation models, 420 certified argmax cases (72 exact ties), 360 races, 300 acceptance decisions with adversarial uniforms, witnesses that each envelope term is necessary, and the R-bf16 and split- K witnesses of Section 3.4 (`tests/test_precision.py`, `evidence/precision/`). These are tests on synthetic heads, not proofs and not GPU measurements.

Formal status. The earlier revision’s Lean file (`formal/DecisionGuards.lean`, nine declarations for scaled-integer guards, monotone refinement, composition, replay and publication guards) could not be compiled when it was written. Lean 4.19.0 now elaborates it, together with `formal/CertifiedArgmax.lean`, which proves the certificate and its uniqueness, the candidate filter, the guards with enclosed numerators and their exclusivity, the residual interval, the Hölder bound and the sequential γ_n bound in scaled integers, using only Lean’s built-in axioms (`formal/STATUS.md`, `evidence/precision/lean_check.log`). IEEE rounding itself, the bridge from floating-point values to these integers, exponentials and every probability statement remain outside the checker.

E Reproduction

The CPU evidence on main is reproduced from the repository root with Python 3.13 and NumPy:

```
python -m venv .venv && . .venv/bin/activate
python -m pip install -r requirements-cpu.txt
python scripts/verify_artifact.py      # CPU tests, synthetic drift, Lean attempt
```

The paper builds with `latexmk -pdf paper.tex` in `paper/` (pdfLaTeX and BibTeX; no shell escape). Figures that read committed data files regenerate on each build. GPU experiments go through the repository’s GPU lock and are documented with their exact commands in `RUNBOOK.md` and in the `README.md` of each evidence/ directory.

F Claim ledger

Claim	Status	Evidence
Head is 15.1% of plain-step weight bytes, 84.1% of an MTP draft step’s	Derived	Checkpoint safetensors headers (Table 1)
GDN state 50.3 MB per request (FP32)	Derived	<code>config.json</code> (Table 1)
Stock head, sampling and verification behaviour at the pin	Code reading	Section 3.2; SGLang bd66ce343e4f
Gap condition (Theorem 3.5) and R-bf16 witness	Proved	Section 3.4
Certified tokens equal stock tokens at the served shapes	Pending	kernel; <code>evidence/certified_head/</code>
Commit rule, directional margin, dead/deferred/enclosed accounting	Proved / definition	Section 3
Transport theorem, summary merge, admission rule	Proved	Section 5
Transport–self-evidence crossover and static-screen comparison	Proved	Theorem 5.5, Proposition 5.6
δ^* median 0.85% (int8, ℓ_2 , 64-row tiles), 0.30% (ℓ_∞/ℓ_1)	Derived	<code>evidence/precision/head_constants.json</code>
Measured drift ρ and transport certification	Pending	geometry; <code>evidence/head_geometry/</code>
Envelope with Hopper accumulation model; integer pass; outward rounding	Proved (model assumed)	Section 6; Khattak and Mikaitis [28]
Refinement ladder; candidate-count bounds	Proved	Theorem 6.6, Proposition 6.7
Candidate sets and head-path runtime on real states	Pending	geometry, kernel
Certified race; RNG contract; fixed-noise verification; coupling bounds	Proved	Section 7

Claim	Status	Evidence
Sampling contract: fixed-noise verification with the stock seeded noise; token-identical to seeded non-speculative decoding	Design decided; pending	kernel; Section 7.3
Common mass, bounded-range sampler, KL and sequence TV bounds	Proved	Propositions 7.3–7.5
Acceptance guards with enclosures; residual races	Proved	Theorems 7.8, 7.9, 7.10
Cheap decision is not cheap verification (constructed heads)	Finding (CPU)	data/synthetic_drift.csv, evidence/synthetic_drift.json
Exact references (inherited)	Finding (CPU)	evidence/decision_tests.json, evidence/race_tests.json
Exact floating-point reference; Lean elaboration	Finding (CPU, Lean)	evidence/precision/, formal/STATUS.md
Attribution, bytes per step, host gaps (H1, H6)	Pending	profile; evidence/profiles/ (PR #13)
Stock divergence mechanisms and state safety (H5)	Pending	state; evidence/state/
DFlash on GH200	Pending	drafter; evidence/drafter/
Certified tail (P1), long-window repair (P2), anchored residual decoding (P3)	Hypotheses	Kill tests pending
What an exact recurrent state must retain (Proposition 3.7); key-subspace reduction	Proved	Section 3.9
State-structure witnesses (−3/25, reachability, SiLU rank, batch rank, RMSNorm, lazy decay, BF16 rank)	Finding (CPU)	evidence/state_structure/
Write avoidance for the recurrent state (P4); cross-request sharing (P5)	Pending; P5 rejected in analysis	moonshot, profile
Stack levers, interaction matrix, exact and lossy frontiers (H4, H7)	Pending	moonshot, bench; evidence/moonshot/, evidence/bench/

G Attribution and disclosure

Samuel Mausberg is the author of this paper and reviews its claims and implementation. The earlier revisions of the manuscript, its derivations and its reference code were prepared with GPT-6 Astra Pro (OpenAI) from research materials, public primary sources and local CPU execution. This revision was produced with Claude Code, using Claude Opus 5.5 agents for the literature review, theory, measurements and writing, on the earlier draft; each agent’s work was reviewed by an independent reviewing agent before it was merged, and all of it is subject to the author’s review. No AI system is an author. The confidential assignment that motivated the work is not reproduced or quoted, and no model weights are redistributed.

References

- [1] Ahmad Abdelfattah, Jack Dongarra, Massimiliano Fasi, Mantas Mikaitis, and Françoise Tisseur. Analysis of Floating-Point Matrix Multiplication Computed via Integer Arithmetic, 2025. URL <https://arxiv.org/abs/2506.11277>. arXiv:2506.11277v3, 8 May 2026 (v1 12 Jun 2025).
- [2] Firas Abuzaid, Geet Sethi, Peter Bailis, and Matei Zaharia. To Index or Not to Index: Optimizing Exact Maximum Inner Product Search. In *2019 IEEE 35th International Conference on Data Engineering (ICDE)*, pages 1250–1261, 2019. doi: 10.1109/ICDE.2019.00114. URL <https://arxiv.org/abs/1706.01449>. arXiv:1706.01449v3, 15 Mar 2019 (v1 5 Jun 2017).
- [3] Ofir Ben Shoham. Balancing Coverage and Draft Latency in Vocabulary Trimming for Faster Speculative Decoding, 2026. URL <https://arxiv.org/abs/2603.05210>. arXiv:2603.05210v1, 5 Mar 2026.
- [4] Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa: Simple LLM Inference Acceleration Framework with Multiple Decoding Heads. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 5209–5235. PMLR, 2024. URL <https://arxiv.org/abs/2401.10774>. arXiv:2401.10774v3, 14 Jun 2024 (v1 19 Jan 2024).
- [5] Yuxuan Cai, Xiaozhuan Liang, Xinghua Wang, Jin Ma, Haijin Liang, Jinwen Luo, Xinyu Zuo, Lisheng Duan, Yuyang Yin, and Xi Chen. FastMTP: Accelerating LLM Inference with Enhanced Multi-Token Prediction, 2025. URL <https://arxiv.org/abs/2509.18362>. arXiv:2509.18362v1, 16 Sep 2025.
- [6] Luis Chamberlain. Certified LM-head decode: read a quarter of the output head, losslessly. `mcgrof/knlp`, `docs/lm-head-decode.md`, first committed 23 June 2026 (commit `1dd92463092d`), 2026. URL <https://github.com/mcgrof/knlp/blob/main/docs/lm-head-decode.md>. MIT licence. Read 30 September 2026.
- [7] Charlie Chen, Sebastian Borgeaud, Geoffrey Irving, Jean-Baptiste Lespiau, Laurent Sifre, and John Jumper. Accelerating Large Language Model Decoding with Speculative Sampling, 2023. URL <https://arxiv.org/abs/2302.01318>. arXiv:2302.01318v1, 2 Feb 2023.
- [8] Jian Chen, Yesheng Liang, and Zhijian Liu. DFlash: Block diffusion for flash speculative decoding. In *International Conference on Machine Learning (ICML)*, 2026. URL <https://arxiv.org/abs/2602.06036v2>. arXiv:2602.06036v2, 28 May 2026 (camera-ready). Code: <https://github.com/z-lab/dflash>.
- [9] Zhiyang Chen, Daliang Xu, Yinyuan Zhang, Chenchua Wang, Mengwei Xu, and Yun Ma. NanoSpec: Accelerating speculative decoding using minimalist in-context vocabularies, 2026. URL <https://arxiv.org/abs/2605.26444v2>. arXiv:2605.26444v2, 1 June 2026. Version 1 (8 April 2026) was titled “MicroSpec: Accelerating Speculative Decoding with Lightweight In-Context Vocabularies”.
- [10] Zhuoming Chen, Avner May, Ruslan Svirschevski, Yuhsun Huang, Max Ryabinin, Zhihao Jia, and Beidi Chen. Sequoia: Scalable, Robust, and Hardware-aware Speculative Decoding. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/ea1f5f0878d43ff4fb8bf64ef4a2326c-Abstract-Conference.html. arXiv:2402.12374v3, 5 Jul 2025 (v1 19 Feb 2024). Proceedings title: “Sequoia: Scalable and Robust Speculative Decoding”.
- [11] Xin Cheng, Xingkai Yu, Chenze Shao, Jiashi Li, Yunfan Xiong, Yi Qian, Jiaqi Zhu, Shirong Ma, Xiaokang Zhang, Jiasheng Ye, Qinyu Chen, Chengqi Deng, Jiping Yu, Damai Dai, Zhengyan Zhang, Yixuan Wei, Yixuan Tan, Wenkai Yang, Runxin Xu, Yu Wu, Zhean Xu, Xuanyu Wang, Muyang Chen, Rui Tian, Xiao Bi, Zhewen Hao, Shaoyuan Chen, Huanqi Cao, Wentao Zhang, Anyi Xu, Huishuai Zhang, Dongyan Zhao, and Wenfeng Liang. DSpark: Confidence-scheduled speculative decoding with semi-autoregressive generation, 2026. URL <https://arxiv.org/abs/2607.05147v1>. arXiv:2607.05147v1, 6 July 2026.
- [12] Majid Daliri, Christopher Musco, and Ananda Theertha Suresh. Coupling without Communication and Drafter-Invariant Speculative Decoding. In *2025 IEEE International Symposium on Information Theory (ISIT)*, pages 1–6, 2025. doi: 10.1109/ISIT63088.2025.11195663. URL <https://arxiv.org/abs/2408.07978>. arXiv:2408.07978v4, 20 Aug 2025 (v1 15 Aug 2024).

- [13] DeepSeek-AI. DeepSeek-V3 Technical Report, 2024. URL <https://arxiv.org/abs/2412.19437>. arXiv:2412.19437v2, 18 Feb 2025 (v1 27 Dec 2024). The arXiv record lists DeepSeek-AI and about 200 named authors.
- [14] Gaoyuan Du, Anam Nawaz Khan, Rex Zhou, Xiaoyang Liu, Deepayan Chakrabarti, Fnu Suyu, and Xueping Li. Greedy decoding is not precision-invariant: Cross-precision output divergence in LLM inference. *Transactions on Machine Learning Research*, 2026. URL <https://arxiv.org/abs/2609.26621v1>. arXiv:2609.26621v1, 22 September 2026; accepted by TMLR per the arXiv comment.
- [15] Massimiliano Fasi, Nicholas J. Higham, Mantas Mikaitis, and Srikara Pranesh. Numerical behavior of NVIDIA tensor cores. *PeerJ Computer Science*, 7:e330, 2021. doi: 10.7717/peerj-cs.330.
- [16] Ophir Frieder, Ida Mele, Cristina Ioana Muntean, Franco Maria Nardini, Raffaele Perego, and Nicola Tonellotto. Caching Historical Embeddings in Conversational Search. *ACM Transactions on the Web*, 18(4), 2024. doi: 10.1145/3578519. URL <https://arxiv.org/abs/2211.14155>. arXiv:2211.14155v1, 25 Nov 2022.
- [17] Yichao Fu, Peter Bailis, Ion Stoica, and Hao Zhang. Break the Sequential Dependency of LLM Inference Using Lookahead Decoding. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 14060–14079. PMLR, 2024. URL <https://arxiv.org/abs/2402.02057>. arXiv:2402.02057v1, 3 Feb 2024.
- [18] Sneha Murthy Ghantasala. TreeWY: Speculative Verification for Gated DeltaNet Hybrids, 2026. URL <https://arxiv.org/abs/2608.20961>. arXiv:2608.20961v1, 21 Aug 2026.
- [19] Fabian Gloeckle, Badr Youbi Idrissi, Baptiste Rozière, David Lopez-Paz, and Gabriel Synnaeve. Better & Faster Large Language Models via Multi-token Prediction. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 15706–15734. PMLR, 2024. URL <https://arxiv.org/abs/2404.19737>. arXiv:2404.19737v1, 30 Apr 2024.
- [20] Raghavv Goel, Sudhanshu Agrawal, Mukul Gagrani, Junyoung Park, Yifan Zao, He Zhang, Tian Liu, Yiping Yang, Xin Yuan, Jiuyan Lu, Chris Lott, and Mingu Lee. VOCABTRIM: Vocabulary Pruning for Efficient Speculative Decoding in LLMs, 2025. URL <https://arxiv.org/abs/2506.22694>. arXiv:2506.22694v2, 3 Jul 2025 (v1 28 Jun 2025). ICML 2025 Workshop on Efficient Systems for Foundation Models, according to the arXiv comment.
- [21] E. J. Gumbel. *Statistical Theory of Extreme Values and Some Practical Applications*. Number 33 in National Bureau of Standards Applied Mathematics Series. National Bureau of Standards, 1954. Bibliographic record checked; the text was not read.
- [22] Stephen Hawkins. Head bit-plane exact argmax: read 25–33 % fewer head bytes on every draft/greedy head read, all families. GitHub issue 69, HawkBearPig/dgpp, opened 29 September 2026; implementation in `src/kernels/packq_head.cu`, 2026. URL <https://github.com/HawkBearPig/dgpp/issues/69>. Apache-2.0. Issue, comments and code read 30 September 2026 at commit a23d8b1. The issue is filed by GitHub user HawkBearPig; the commit author is Stephen Hawkins.
- [23] Horace He and Thinking Machines Lab. Defeating nondeterminism in LLM inference. Thinking Machines Lab: Connectionism, September 2025. URL <https://thinkingmachines.ai/blog/defeating-nondeterminism-in-llm-inference/>. Published 10 September 2025. Code: https://github.com/thinking-machines-lab/batch_invariant_ops.
- [24] Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. Society for Industrial and Applied Mathematics, Philadelphia, PA, second edition, 2002. doi: 10.1137/1.9780898718027.
- [25] Kaixuan Huang, Xudong Guo, and Mengdi Wang. SpecDec++: Boosting Speculative Decoding via Adaptive Candidate Lengths. In *Second Conference on Language Modeling (COLM)*, 2025. URL <https://arxiv.org/abs/2405.19715>. arXiv:2405.19715v3, 11 Jul 2025 (v1 30 May 2024). Venue from the arXiv comment.
- [26] Inco AI. DFlash 2: Keep drafting parallel. Inco AI blog, August 2026. URL <https://inco.ai/blog/dflash2/>. Published 18 August 2026; read 30 September 2026.

- [27] Inco AI. Qwen3.8-27B-DFlash2 model card. Hugging Face, revision `015e795645c74b1a0eeef3b570031fb62e769bc5`, 2026. URL <https://huggingface.co/incoai/Qwen3.8-27B-DFlash2/tree/015e795645c74b1a0eeef3b570031fb62e769bc5>. Created 18 August 2026; read 30 September 2026.
- [28] Faizan A. Khattak and Mantas Mikaitis. Accurate Models of NVIDIA Tensor Cores. *ACM Transactions on Architecture and Code Optimization*, 23(3):1–25, 2026. doi: 10.1145/3830409. URL <https://arxiv.org/abs/2512.07004>. arXiv:2512.07004v4, 11 Jun 2026 (v1 7 Dec 2025).
- [29] Şuayp Talha Kocabay, Talha Rüzgar Akkuş, and Kamer Ali Yuksel. ARCHead: Activation-Metric Residual Correction for Large Language Model Output Heads, 2026. URL <https://arxiv.org/abs/2608.02703>. arXiv:2608.02703v1, 3 Aug 2026. Submitted to ACL Rolling Review.
- [30] Wouter Kool, Herke van Hoof, and Max Welling. Stochastic Beams and Where to Find Them: The Gumbel-Top-k Trick for Sampling Sequences Without Replacement. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3499–3508. PMLR, 2019. URL <https://arxiv.org/abs/1903.06059>. arXiv:1903.06059v2, 29 May 2019 (v1 14 Mar 2019).
- [31] Siqi Kou, Lanxiang Hu, Zhezhi He, Zhijie Deng, and Hao Zhang. CLLMs: Consistency Large Language Models, 2024. URL <https://arxiv.org/abs/2403.00835>. arXiv:2403.00835v4, 13 Jun 2024 (v1 28 Feb 2024). ICML 2024 according to the arXiv comment.
- [32] Yashas Samaga B L, Varun Yerram, Chong You, Srinadh Bhojanapalli, Sanjiv Kumar, Prateek Jain, and Praneeth Netrapalli. HiRE: High Recall Approximate Top-k Estimation for Efficient LLM Inference, 2024. URL <https://arxiv.org/abs/2402.09360>. arXiv:2402.09360v1, 14 Feb 2024.
- [33] Layr Labs. Laguna LM-head prune (Sources/MLXFastModel/LagunaLmHeadPrune.swift). Source code, Layr-Labs/mlxfast-challenge, commit 4f4f68959cfe (28 July 2026), 2026. URL <https://github.com/Layr-Labs/mlxfast-challenge>. MIT licence. Read 30 September 2026.
- [34] Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 19274–19286. PMLR, 2023. URL <https://proceedings.mlr.press/v202/leviathan23a.html>. arXiv:2211.17192.
- [35] Hui Li, Tsz Nam Chan, Man Lung Yiu, and Nikos Mamoulis. FEXIPRO: Fast and exact inner product retrieval in recommender systems. In *Proceedings of the 2017 ACM International Conference on Management of Data (SIGMOD)*, pages 835–850, 2017. doi: 10.1145/3035918.3064009.
- [36] Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE: Speculative Sampling Requires Rethinking Feature Uncertainty. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 28935–28948. PMLR, 2024. URL <https://arxiv.org/abs/2401.15077>. arXiv:2401.15077v3, 4 Mar 2025 (v1 26 Jan 2024).
- [37] Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE-2: Faster Inference of Language Models with Dynamic Draft Trees. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 7421–7432, 2024. doi: 10.18653/v1/2024.emnlp-main.422. URL <https://arxiv.org/abs/2406.16858>. arXiv:2406.16858v2, 30 Jun 2024 (v1 24 Jun 2024).
- [38] Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE-3: Scaling up Inference Acceleration of Large Language Models via Training-Time Test. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 2025. URL <https://arxiv.org/abs/2503.01840>. arXiv:2503.01840v3, 23 Apr 2025 (v1 3 Mar 2025). <https://neurips.cc/virtual/2025/poster/119930>.
- [39] Ze-Wei Liou and Tri Dao. ReplaySSM: Cache SSM inputs, not state. Dao AI Lab blog, June 2026. URL <https://dao-lab.ai/blog/2026/replayssm/>. Published 15 June 2026. Code: <https://github.com/Johnny-Liou/ReplaySSM>.

- [40] Dong Liu, Shu Wang, Yanxuan Yu, Haisheng Wang, and Ben Lengerich. CSV-Decode: Certifiable Sub-Vocabulary Decoding for Efficient Large Language Model Inference, 2025. URL <https://arxiv.org/abs/2511.21702>. arXiv:2511.21702v2, 26 Jul 2026 (v1 16 Nov 2025). Code: <https://github.com/FastLM/CSV-Decode> (no licence file).
- [41] Tianyu Liu, Yuhao Shen, Rui Cen, Junhan Shi, Jiebin Zhang, Guangshuo Qin, Hong Liu, Song Liu, Guanghua Yu, and Jianchen Zhu. D-cut: Adaptive verification depth pruning for batched speculative decoding, 2026. URL <https://arxiv.org/abs/2607.14647v1>. arXiv:2607.14647v1, 16 July 2026.
- [42] Zhiyuan Ma. GDN Tree-Scan: Served Tree Verification for Recurrent-Hybrid Language Models, 2026. URL <https://arxiv.org/abs/2609.23900>. arXiv:2609.23900v1, 20 Sep 2026.
- [43] Chris J. Maddison, Daniel Tarlow, and Tom Minka. A* sampling. In *Advances in Neural Information Processing Systems 27 (NIPS 2014)*, pages 3086–3094, 2014. URL https://proceedings.neurips.cc/paper_files/paper/2014/hash/937debc749f041eb5700df7211ac795c-Abstract.html. arXiv:1411.0030.
- [44] Stephen Mussmann and Stefano Ermon. Learning and inference via maximum inner product search. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 2587–2596. PMLR, 2016. URL <https://proceedings.mlr.press/v48/mussmann16.html>.
- [45] Stephen Mussmann, Daniel Levy, and Stefano Ermon. Fast amortized inference and learning in log-linear models with randomly perturbed nearest neighbor search. In *Proceedings of the Thirty-Third Conference on Uncertainty in Artificial Intelligence (UAI 2017)*, 2017. URL <https://arxiv.org/abs/1707.03372>. arXiv:1707.03372.
- [46] NVIDIA. NVIDIA Hopper tuning guide. CUDA 13.0.2 documentation archive, 2025. URL <https://docs.nvidia.com/cuda/archive/13.0.2/hopper-tuning-guide/index.html>. Read 30 September 2026.
- [47] NVIDIA. AIPerf metrics reference. AIPerf documentation, 2026. URL <https://docs.nvidia.com/aiperf/reference/ai-perf-metrics-reference>. Read 30 September 2026.
- [48] NVIDIA. NVIDIA Blackwell tuning guide. CUDA 13.4 documentation, 2026. URL <https://docs.nvidia.com/cuda/blackwell-tuning-guide/index.html>. Read 30 September 2026.
- [49] NVIDIA. Floating point and IEEE 754. CUDA 13.4 documentation, 2026. URL <https://docs.nvidia.com/cuda/floating-point/index.html>. Read 30 September 2026.
- [50] NVIDIA. Nsight Compute profiling guide. Nsight Compute documentation, 2026. URL <https://docs.nvidia.com/nsight-compute/ProfilingGuide/index.html>. Read 30 September 2026.
- [51] NVIDIA. Parallel thread execution ISA, version 9.4. CUDA documentation, 2026. URL <https://docs.nvidia.com/cuda/parallel-thread-execution/index.html>. Sec. 9.7.17 (warpgroup MMA, requires sm_90a) and Sec. 9.7.18 (tcgen05); read 30 September 2026.
- [52] NVIDIA CUTLASS. CuTe DSL overview. CUTLASS documentation, 2026. URL <https://docs.nvidia.com/cutlass/latest/media/docs/pythonDSL/overview.html>. Read 30 September 2026.
- [53] Yi Pan, Haocheng Xi, Kan Zhu, Xingyang Li, Yibo Wu, Mayank Mishra, Hongtao Zhang, William X. Zheng, Baris Kasikci, Song Han, Kurt Keutzer, Rishabh Iyer, and Ion Stoica. LeapQuant: Efficient Linear Attention with Accurate Recurrent State Quantization, 2026. URL <https://arxiv.org/abs/2609.38166>. arXiv:2609.38166v1, 29 Sep 2026.
- [54] George Papandreou and Alan L. Yuille. Perturb-and-MAP random fields: Using discrete optimization to learn and sample from energy models. In *2011 International Conference on Computer Vision (ICCV)*, pages 193–200, 2011. doi: 10.1109/ICCV.2011.6126242.
- [55] Pragaash Ponnusamy, Shivam Sahni, Jue Wang, and Tri Dao. SonicSampler: Unified tile-aware kernels for LLM sampling and speculative verification, 2026. URL <https://arxiv.org/abs/2607.20475v1>. arXiv:2607.20475v1, submitted 24 May 2026.

- [56] PyTorch contributors. Numerical accuracy. PyTorch 2.14 documentation, 2026. URL https://docs.pytorch.org/docs/2.14/notes/numerical_accuracy.html. Read 30 September 2026.
- [57] Qwen. Qwen3.5-4B model card. Hugging Face, revision 851bf6e806efd8d0a36b00ddf55e13ccb7b8cd0a, 2026. URL <https://huggingface.co/Qwen/Qwen3.5-4B/tree/851bf6e806efd8d0a36b00ddf55e13ccb7b8cd0a>. Read 30 September 2026.
- [58] Qwen. Qwen3.5-4B model configuration (config.json). Hugging Face, revision 851bf6e806efd8d0a36b00ddf55e13ccb7b8cd0a, 2026. URL <https://huggingface.co/Qwen/Qwen3.5-4B/blob/851bf6e806efd8d0a36b00ddf55e13ccb7b8cd0a/config.json>. Read 30 September 2026.
- [59] qwen38-challenge_senpai contributors. E142: The verify readout streams 715 MB per round to answer a two-element question. GitHub pull request 142, morganmcg1/qwen38-challenge_senpai, opened 22 August 2026, 2026. URL https://github.com/morganmcg1/qwen38-challenge_senpai/pull/142. Final report comment of 23 August 2026 records the screens as refuted. Read 30 September 2026.
- [60] Parikshit Ram and Alexander G. Gray. Maximum Inner-Product Search Using Cone Trees. In *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 931–939, 2012. doi: 10.1145/2339530.2339677. URL <https://arxiv.org/abs/1202.6101>. arXiv:1202.6101v1, 28 Feb 2012. arXiv title: “Maximum Inner-Product Search using Tree Data-structures”.
- [61] Joseph Rowan, Buu Phan, and Ashish Khisti. List-Level Distribution Coupling with Applications to Speculative Decoding and Lossy Compression. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 2025. URL <https://arxiv.org/abs/2506.05632>. arXiv:2506.05632v3, 11 Jan 2026 (v1 5 Jun 2025).
- [62] Tomas Ruiz, Zhen Qin, Yifan Zhang, Xuyang Shen, Yiran Zhong, and Mengdi Wang. FlashSampling: Fast and memory-efficient exact sampling, 2026. URL <https://arxiv.org/abs/2603.15854v3>. arXiv:2603.15854v3, 25 September 2026 (v1 16 March 2026, v2 12 May 2026). Code: <https://github.com/FlashSampling/FlashSampling>.
- [63] Siegfried M. Rump. Verification methods: Rigorous results using floating-point arithmetic. *Acta Numerica*, 19: 287–449, 2010. doi: 10.1017/S096249291000005X.
- [64] Matan Rusanovsky, Yoav Miron, Roy Uziel, Omer Belhasin, Hao Guo, Ran Zilberstein, Maor Ashkenazi, and Michael Elad. LiLiCorr: Lightweight likelihood correlation of parallel drafts for speculative decoding, 2026. URL <https://arxiv.org/abs/2608.20530v2>. arXiv:2608.20530v2, 22 September 2026.
- [65] SGLang contributors. GDN ReplaySSM speculative fold (python/sglang/kernels/ops/attention/fla/gdn_replayssm_spec_fold.py). Source code, commit bd66ce343e4f6e2f2b75d7e820fe4d0718a8d824, 2026. URL https://github.com/sgl-project/sglang/blob/bd66ce343e4f6e2f2b75d7e820fe4d0718a8d824/python/sglang/kernels/ops/attention/fla/gdn_replayssm_spec_fold.py. Lines 1–95 read 30 September 2026.
- [66] SGLang contributors. [feature] dynamic verification for DFlash2. SGLang pull request 36136, opened 24 August 2026 by GitHub user EthanWang555, 2026. URL <https://github.com/sgl-project/sglang/pull/36136>. Open and unmerged on 30 September 2026; head c0235e700214f1de5044be25f795faf5b755e450.
- [67] SGLang contributors. DFlash model and candidate-head implementation (python/sglang/srt/models/dflash.py). Source code, commit bd66ce343e4f6e2f2b75d7e820fe4d0718a8d824, 2026. URL <https://github.com/sgl-project/sglang/blob/bd66ce343e4f6e2f2b75d7e820fe4d0718a8d824/python/sglang/srt/models/dflash.py>. Read 30 September 2026.
- [68] Kyuhong Shim, Minjae Lee, Iksoo Choi, Yoonho Boo, and Wonyong Sung. SVD-Softmax: Fast softmax approximation on large vocabulary neural networks. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/4e2a6330465c8ffc6a696a5a16639176-Abstract.html.

- [69] sparkpipe contributors. Certified screened LM head (SparkLmHeadCertifiedFp8QuantizeKernel, SparkLmHeadCertifiedFp8ScoreKernel). Source code, `model-families/common/include/sparkpipe/spark_lm_kernels.cuh`, commit 84efd5b63d8d (16 August 2026); pull request 744 (merged 29 August 2026), 2026. URL <https://github.com/sparkpipe/sparkpipe/pull/744>. No licence file. Read 30 September 2026.
- [70] Ziteng Sun, Ananda Theertha Suresh, Jae Hun Ro, Ahmad Beirami, Himanshu Jain, and Felix Yu. SpecTr: Fast Speculative Decoding via Optimal Transport. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/6034a661584af6c28fd97a6f23e56c0a-Abstract-Conference.html. arXiv:2310.15141v2, 18 Jan 2024 (v1 23 Oct 2023).
- [71] Ziteng Sun, Uri Mendlovic, Yaniv Leviathan, Asaf Aharoni, Jae Hun Ro, Ahmad Beirami, and Ananda Theertha Suresh. Block Verification Accelerates Speculative Decoding. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL <https://arxiv.org/abs/2403.10444>. arXiv:2403.10444v3, 10 Apr 2025 (v1 15 Mar 2024).
- [72] Chaofan Tao, Qian Liu, Longxu Dou, Niklas Muennighoff, Zhongwei Wan, Ping Luo, Min Lin, and Ngai Wong. Scaling Laws with Vocabulary: Larger Models Deserve Larger Vocabularies. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/cf5a019ae9c11b4be88213ce3f85d85c-Abstract-Conference.html. arXiv:2407.13623v3, 1 Nov 2024 (v1 18 Jul 2024).
- [73] Christina Teflioudi, Rainer Gemulla, and Olga Mykytiuk. LEMP: Fast retrieval of large entries in a matrix product. In *Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data*, pages 107–122, 2015. doi: 10.1145/2723372.2747647.
- [74] The SGLang Team. Towards deterministic inference in SGLang and reproducible RL training. LMSYS Org blog, September 2025. URL <https://www.lmsys.org/blog/2025-09-22-sglang-deterministic/>. Published 22 September 2025, updated 24 September 2025.
- [75] TileLang contributors. TileLang documentation. Version 0.1.15, 2026. URL <https://tilelang.com/>. Read 30 September 2026.
- [76] Triton contributors. `triton.autotune`. Triton documentation, 2026. URL <https://triton-lang.org/main/python-api/generated/triton.autotune.html>. Read 30 September 2026.
- [77] vLLM contributors. CUDA graphs. vLLM design documentation, 2026. URL https://docs.vllm.ai/en/latest/design/cuda_graphs/. Read 30 September 2026.
- [78] Jie Wang, Peter Wonka, and Jieping Ye. Lasso screening rules via dual polytope projection. *Journal of Machine Learning Research*, 16:1063–1101, 2015. URL <https://jmlr.org/papers/v16/wang15a.html>.
- [79] Jikai Wang, Yi Su, Juntao Li, Qingrong Xia, Zi Ye, Xinyu Duan, Zhefeng Wang, and Min Zhang. OPT-Tree: Speculative decoding with adaptive draft tree structure. *Transactions of the Association for Computational Linguistics*, 13:188–199, 2025. doi: 10.1162/tacl_a_00735. URL <https://aclanthology.org/2025.tacl-1.8/>. arXiv:2406.17276.
- [80] Junxiong Wang, Daniele Paliotta, Avner May, Alexander M. Rush, and Tri Dao. The Mamba in the Llama: Distilling and Accelerating Hybrid Models. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/723933067ad315269b620bc0d2c05cba-Abstract-Conference.html. arXiv:2408.15237v4, 27 Jun 2025 (v1 27 Aug 2024).
- [81] Li Wang, Yi Su, Xiabao Wu, Chiran You, Yongchao Liu, Zhan Qiu, Juelu Zhang, Jiajun Zheng, Fangxin Liu, Jie Zhang, Chen Tian, and Chengying Huan. Bole: Efficient Tree Speculation for Hybrid-Attention Language Models, 2026. URL <https://arxiv.org/abs/2608.01651>. arXiv:2608.01651v1, 3 Aug 2026.
- [82] Zhibin Wang, Xuying Han, Zhaohua Yang, Fuliang Liu, Xue Li, Rong Gu, Sheng Zhong, and Chen Tian. SpecLA: Efficient Speculative Decoding for Linear-Attention Models, 2026. URL <https://arxiv.org/abs/2607.16673>. arXiv:2607.16673v1, 18 Jul 2026.

- [83] Roger Weber, Hans-J. Schek, and Stephen Blott. A quantitative analysis and performance study for similarity-search methods in high-dimensional spaces. In *Proceedings of the 24th International Conference on Very Large Data Bases (VLDB 1998)*, 1998. URL <https://www.vldb.org/conf/1998/p194.pdf>.
- [84] Miles Williams, Young D. Kwon, Rui Li, Alexandros Kouris, and Stylianos I. Venieris. Speculative decoding with a speculative vocabulary. In *Findings of the Association for Computational Linguistics: ACL 2026*, 2026. URL <https://arxiv.org/abs/2602.13836v2>. arXiv:2602.13836v2, 17 July 2026.
- [85] Mengdi Wu, Xinhao Cheng, Shengyu Liu, Chunan Shi, Jianan Ji, Man Kit Ao, Praveen Velliengiri, Xupeng Miao, Oded Padon, and Zhihao Jia. Mirage: A multi-level superoptimizer for tensor programs. In *19th USENIX Symposium on Operating Systems Design and Implementation (OSDI 25)*. USENIX Association, 2025. URL <https://www.usenix.org/conference/osdi25/presentation/wu-mengdi>. arXiv:2405.05751v3.
- [86] Yangchao Wu, Zongyue Qin, Alex Wong, and Stefano Soatto. STree: Speculative Tree Decoding for Hybrid State-Space Models. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 2025. URL <https://arxiv.org/abs/2505.14969>. arXiv:2505.14969v2, 27 Oct 2025 (v1 20 May 2025). <https://neurips.cc/virtual/2025/poster/117297>.
- [87] Shanli Xing, Zihao Ye, Bohan Hou, Luis Ceze, and Tianqi Chen. Sorting-free GPU kernels for LLM sampling. FlashInfer blog, March 2025. URL <https://flashinfer.ai/2025/03/10/sampling.html>. Published 10 March 2025.
- [88] Songlin Yang, Bailin Wang, Yu Zhang, Yikang Shen, and Yoon Kim. Parallelizing Linear Transformers with the Delta Rule over Sequence Length. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/d13a3eae72366e61dfdc7eea82eeb685-Abstract-Conference.html. arXiv:2406.06484v6, 15 Jan 2025 (v1 10 Jun 2024).
- [89] Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving Mamba2 with delta rule. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL <https://openreview.net/forum?id=r8H7xhYPwz>. arXiv:2412.06464v3.
- [90] Zihao Ye, Lequn Chen, Ruihang Lai, Wuwei Lin, Yineng Zhang, Stephanie Wang, Tianqi Chen, Baris Kasikci, Vinod Grover, Arvind Krishnamurthy, and Luis Ceze. FlashInfer: Efficient and customizable attention engine for LLM inference serving. In *Proceedings of the Eighth Conference on Machine Learning and Systems (MLSys 2025)*, 2025. URL <https://arxiv.org/abs/2501.01005>. arXiv:2501.01005v2.
- [91] Jiayi Yuan, Hao Li, Xinheng Ding, Wenya Xie, Yu-Jhe Li, Wentian Zhao, Kun Wan, Jing Shi, Xia Hu, and Zirui Liu. Understanding and Mitigating Numerical Sources of Nondeterminism in LLM Inference. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 2025. URL <https://arxiv.org/abs/2506.09501>. arXiv:2506.09501v2, 24 Oct 2025 (v1 11 Jun 2025). Version 1 was titled “Give Me FP32 or Give Me Death? Challenges and Solutions for Reproducible Reasoning”.
- [92] z-lab. Qwen3.5-4B-DFlash model card. Hugging Face, revision 9a1996ccf887b79ab3af4fcbf8c1d1f4b5658bcf, 2026. URL <https://huggingface.co/z-lab/Qwen3.5-4B-DFlash/tree/9a1996ccf887b79ab3af4fcbf8c1d1f4b5658bcf>. Apache-2.0; created 5 March 2026, updated 19 June 2026; mirrored at `moda1-labs/Qwen3.5-4B-DFlash`. DFlash (v1) drafter for Qwen/Qwen3.5-4B with SGLang launch instructions and B200 benchmarks against MTP. Read 30 September 2026.
- [93] Chunran Zhang. Exact Adaptive Hybrid Retrieval Without Fixed Top-L Cutoffs, 2026. URL <https://arxiv.org/abs/2608.07152>. arXiv:2608.07152v1, 7 Aug 2026.
- [94] Jinbin Zhang, Nasib Ullah, Erik Schultheis, and Rohit Babbar. DynaSpec: Context-aware dynamic speculative sampling for large-vocabulary language models, 2025. URL <https://arxiv.org/abs/2510.13847v3>. arXiv:2510.13847v3, 3 February 2026 (v1 11 October 2025).
- [95] Tao Zhang, Jianchao Tan, Pingwei Sun, Yanqi Yu, Zixu Jiang, Yuchen Xie, Xunliang Cai, and Ziqian Zeng. DAMP: Decay-Aware Mixed-Precision Recurrent-State Quantization, 2026. URL <https://arxiv.org/abs/2608.27513>. arXiv:2608.27513v1, 27 Aug 2026.

- [96] Weilin Zhao, Tengyu Pan, Xu Han, Yudi Zhang, Ao Sun, Yuxiang Huang, Kaihuo Zhang, Weilun Zhao, Yuxuan Li, Jie Zhou, Hao Zhou, Jianyong Wang, Zhiyuan Liu, and Maosong Sun. FR-Spec: Accelerating Large-Vocabulary Language Models via Frequency-Ranked Speculative Sampling. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3909–3921, 2025. doi: 10.18653/v1/2025.acl-long.198. URL <https://arxiv.org/abs/2502.14856>. arXiv:2502.14856v2, 11 Mar 2025 (v1 20 Feb 2025).
- [97] Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and Ying Sheng. SGLang: Efficient execution of structured language model programs. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, pages 62557–62583, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/724be4472168f31ba1c9ac630f15dec8-Abstract-Conference.html. arXiv:2312.07104.